Time Series Data Analysis

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Outline

1 Stylized Facts of Financial Returns

2 Efficient Market Hypothesis and Statistical Models for Financial Returns

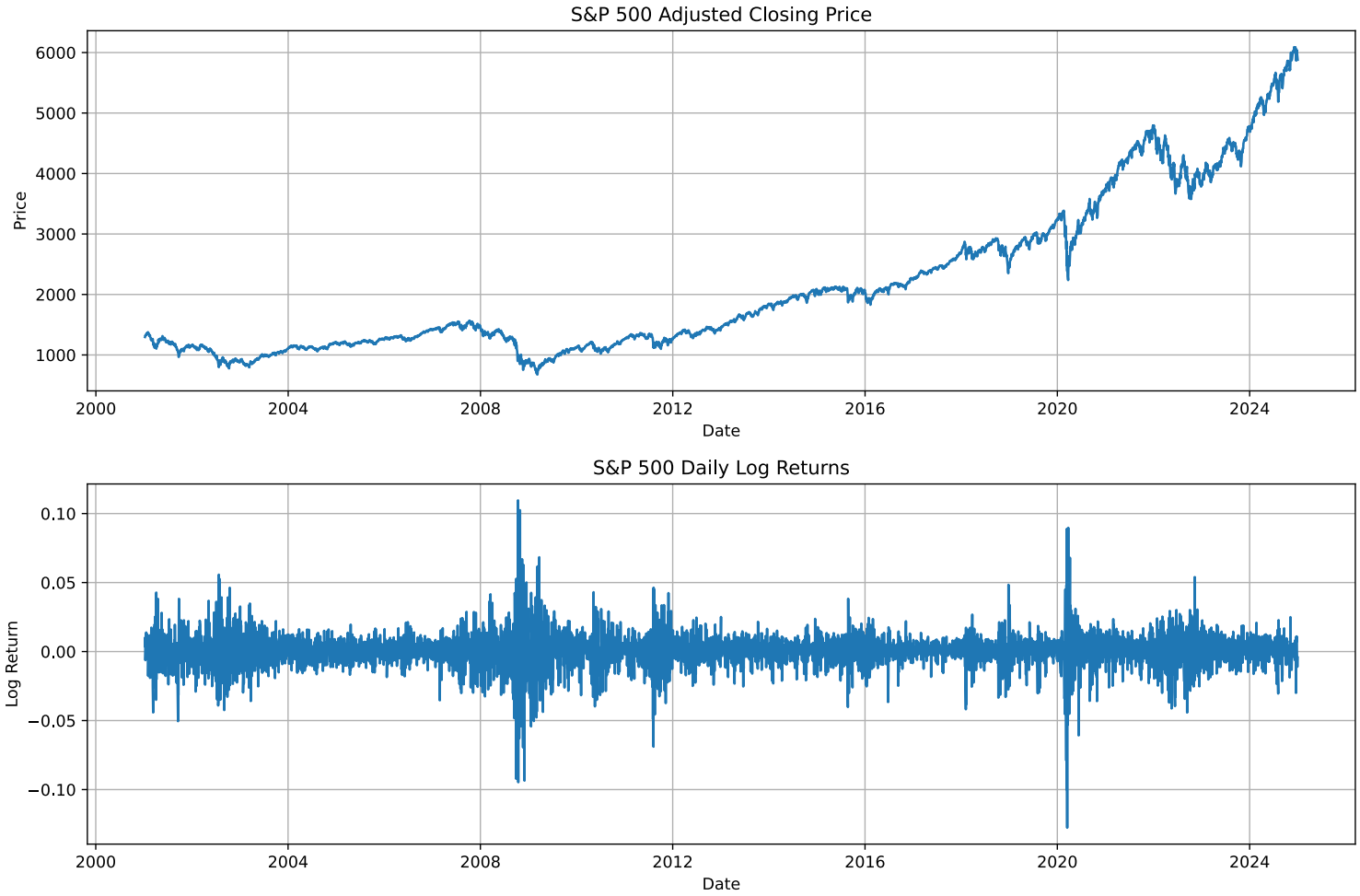
3 Time Series Models for Financial Returns

4 Volatility Modeling via GARCH Models

5 Applications and Extensions of GARCH Models

The Data

We will use the S&P 500 index data (2001-01-08 to 2024-12-31) from Yahoo Finance to illustrate the stylized facts of financial returns.



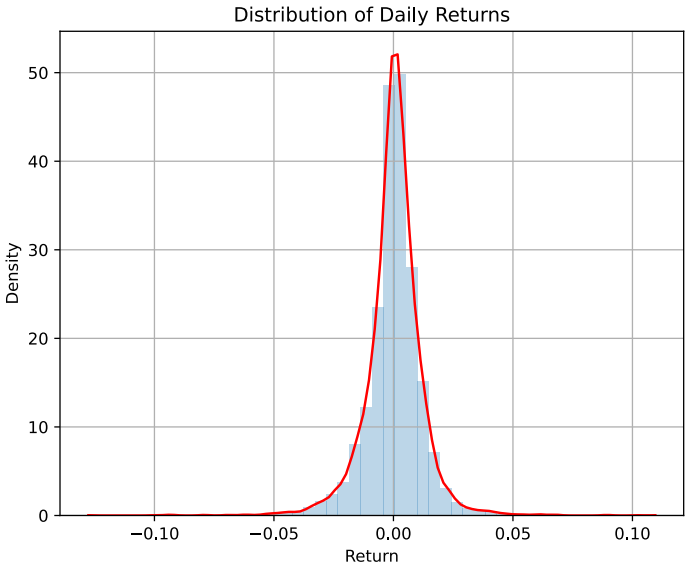
Stationarity

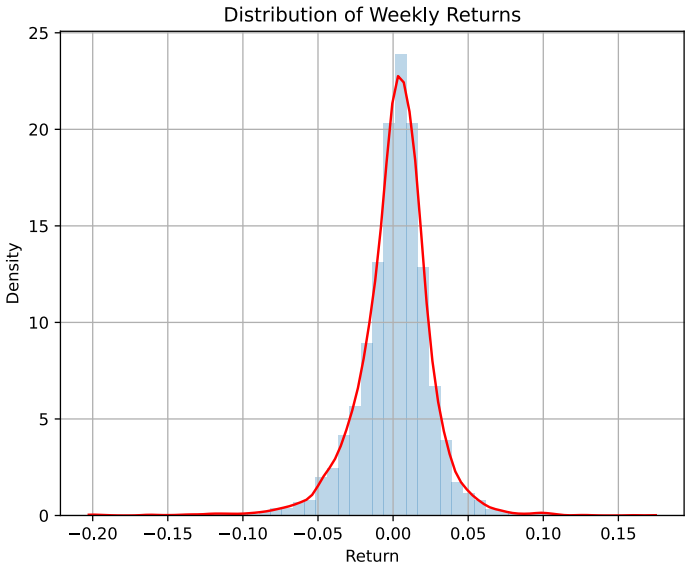
The prices of an asset recorded over time are often not sationary, due to, for example,

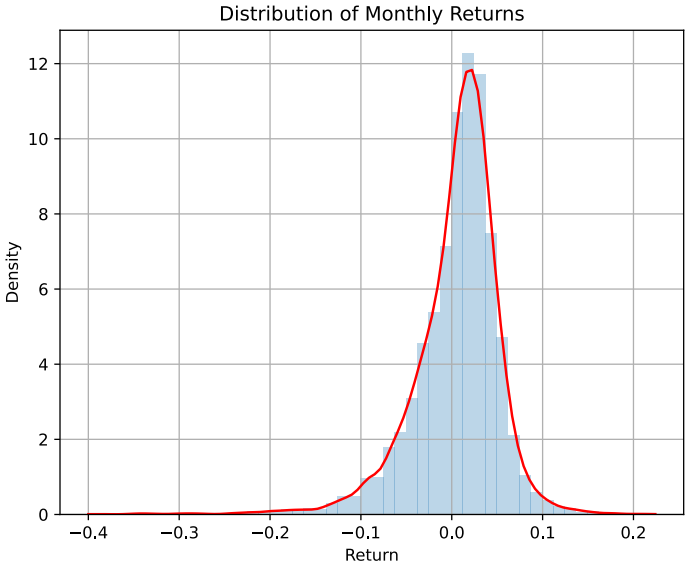
- the steady expansion of an economy,
- the increase of productivity due to technological progress,
- economic recessions and financial crises.

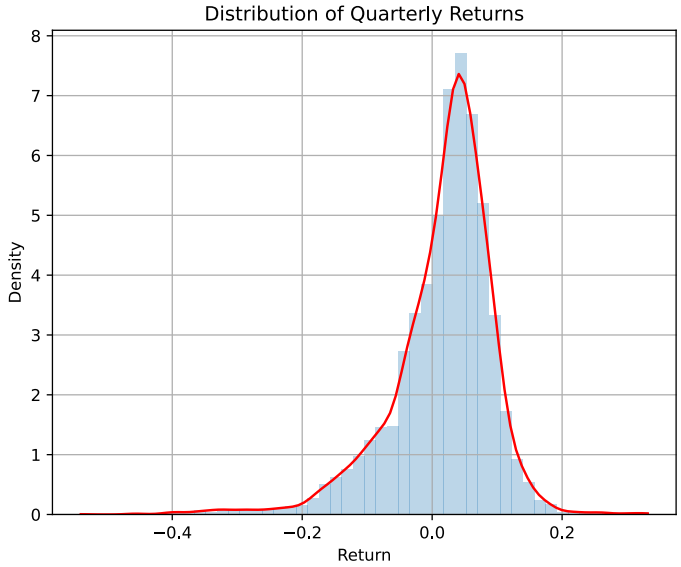
However, the returns are often stationary, fluctuating around a constant level.

Distributional Properties

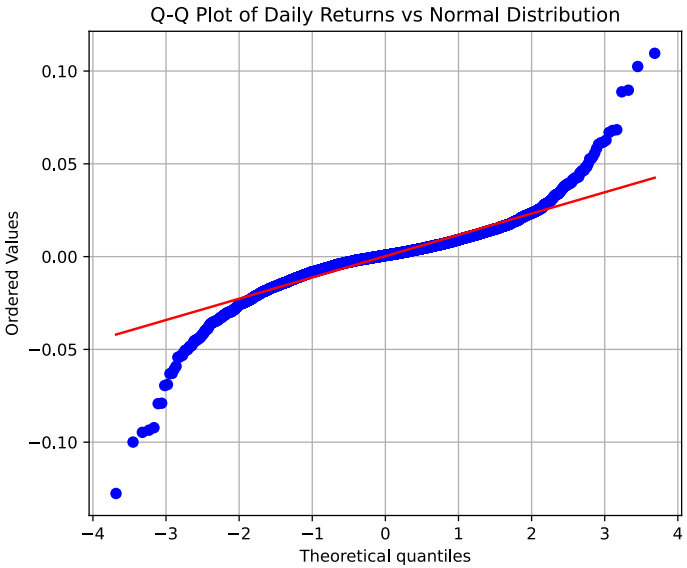


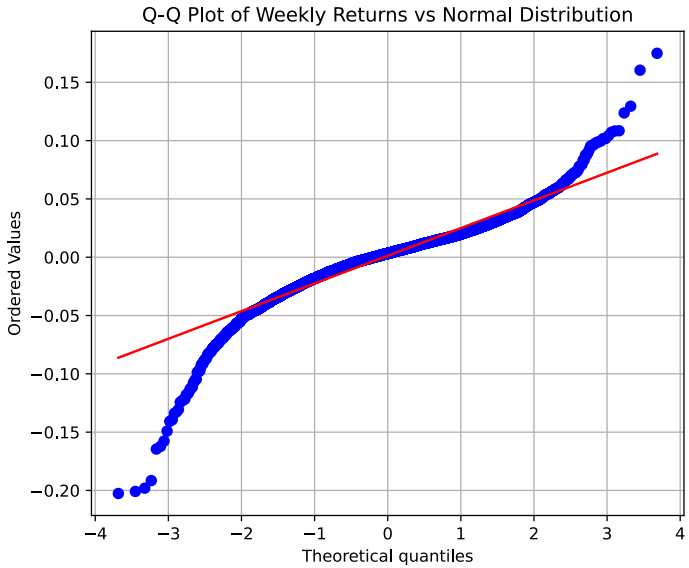


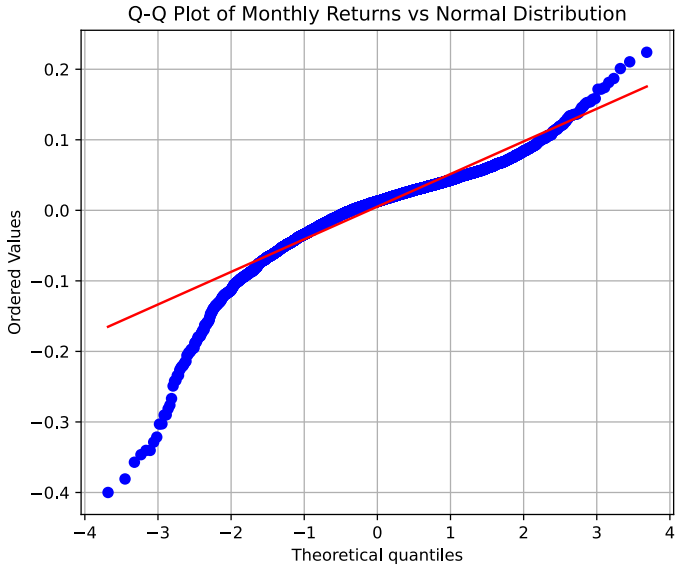


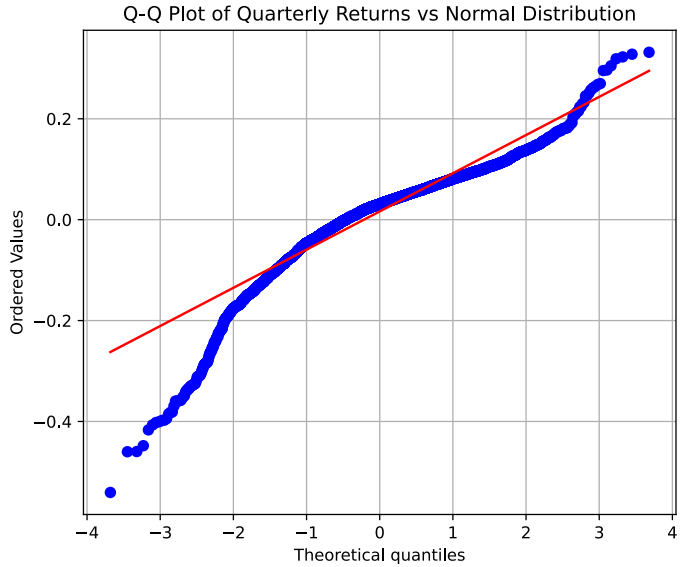


Distributional Properties (continued)







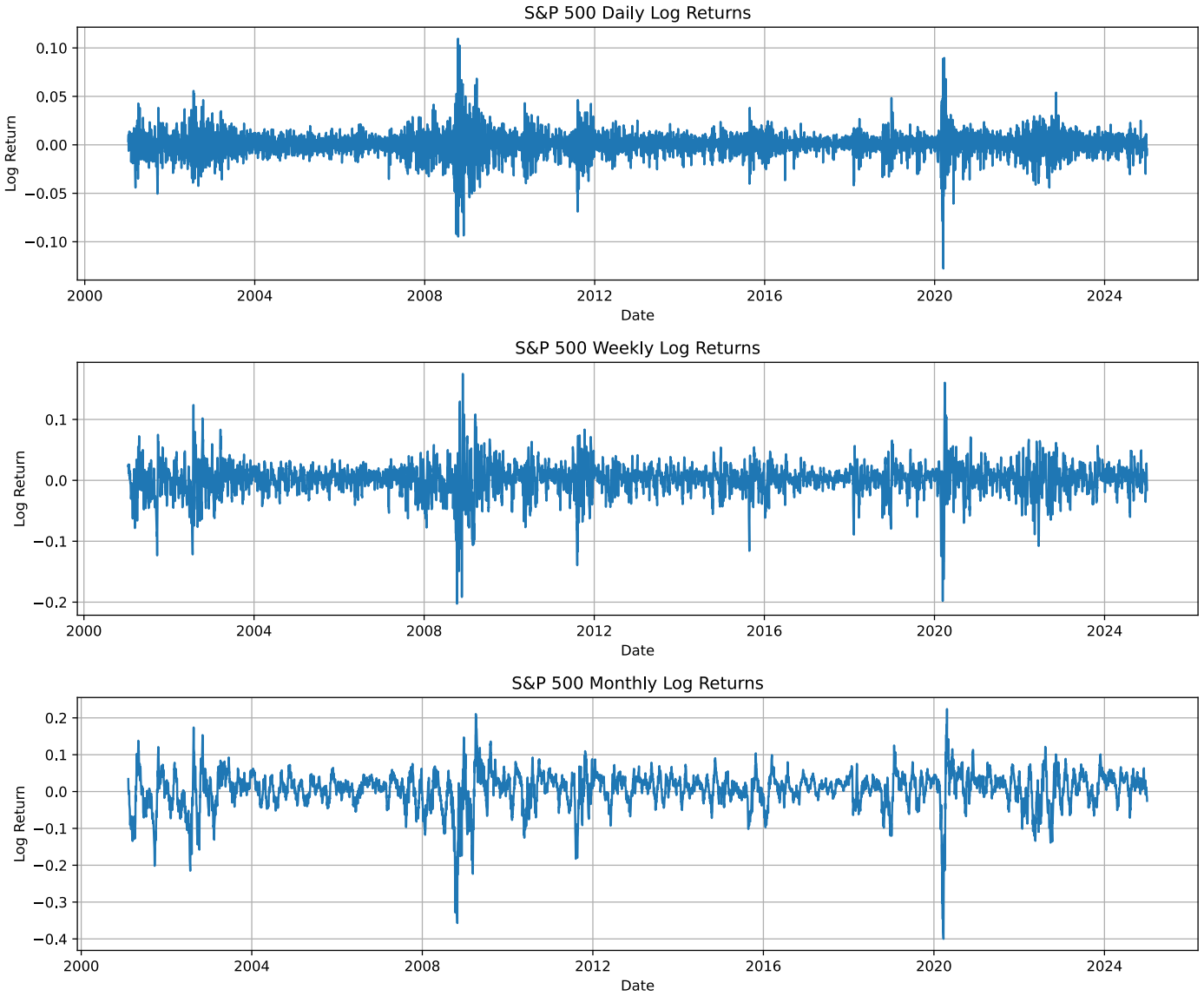


Distributional Properties (continued)

From the QQ plot, we can see that the S&P 500 returns are heavy-tailed and left-skewed.

As the sampling frequency increases, the distribution of returns becomes more and more heavy-tailed, while the left-skewness is quite persistent for various sampling frequencies.

Volatility Clustering



Volatility Clustering (continued)

As the sampling frequency increases, the volatility clustering becomes more and more pronounced.

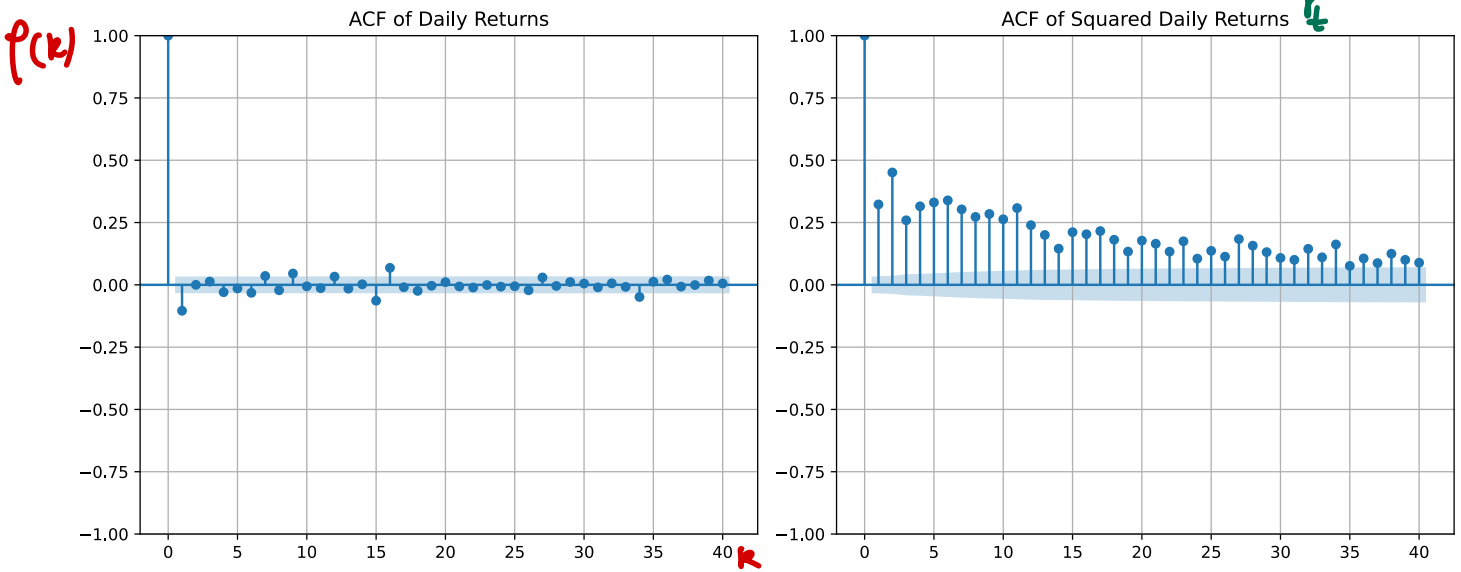
The high volatility periods are followed by high volatility periods, e.g., the financial crisis in 2008.

Long-Range Dependence

The autocorrelation function (ACF) is defined as $\rho(k) = \gamma(k)/\gamma(0)$, where $\gamma(k)$ is the autocovariance function:

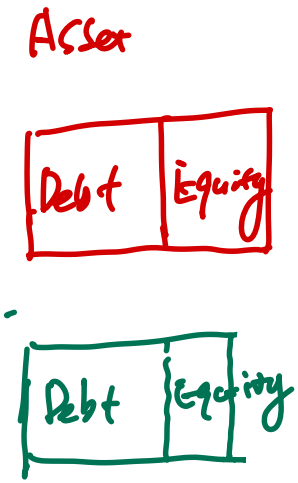
$$\gamma(k) = \mathrm{Cov}(r_t, r_{t-k}) \approx \frac{1}{T-k} \sum_{t=k+1}^{T} (r_t - \bar{r})(r_{t-k} - \bar{r}).$$

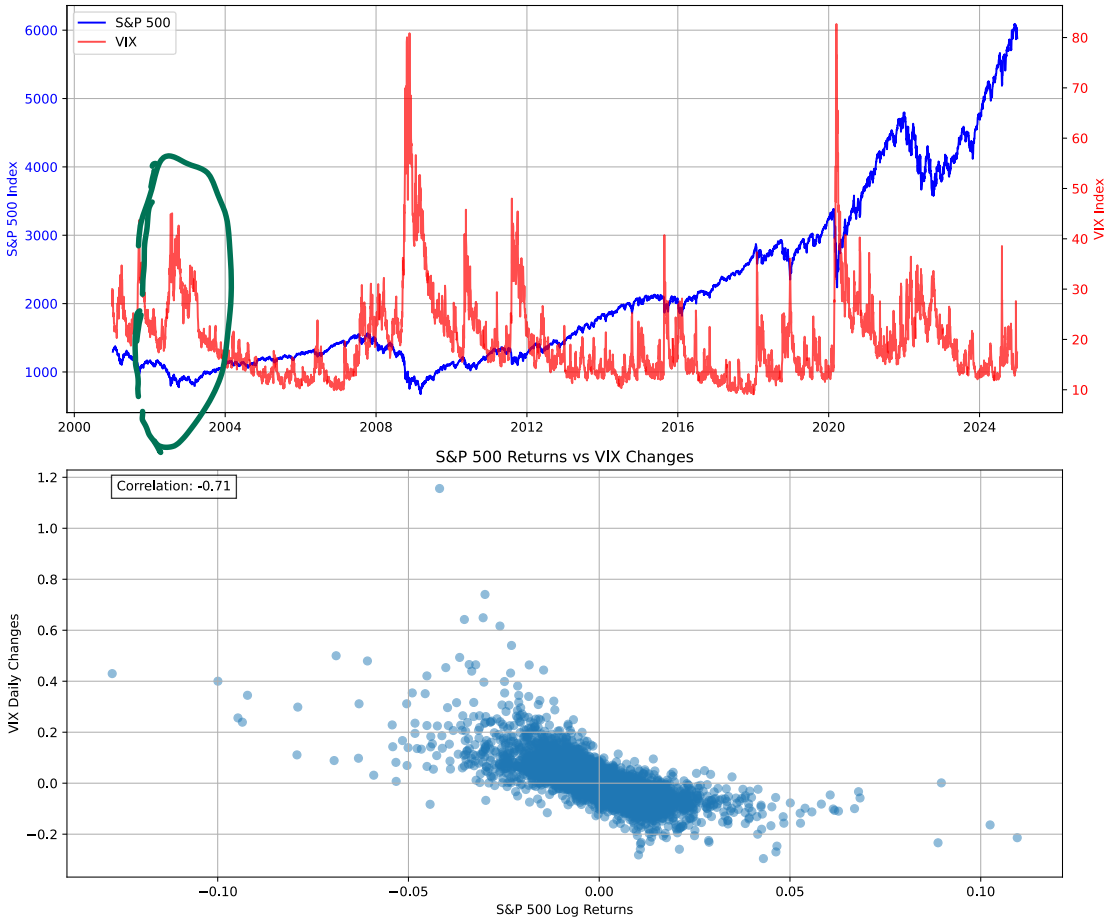
For log returns, $\rho(k) \approx 0$ for $k \geq 1$, while for the squared returns, $\rho(k)$ decays slowly, indicating long-range dependence.



Leverage Effect: VIX and S&P 500 Returns

When the SPX index drops, the firms are more leveraged, and the risk of the firms increases, leading to increasing volatility (proxied by VIX).





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Efficient Market Hypothesis

The efficient market hypothesis (EMH) states that the prices of an asset reflect all available information, and it is impossible to consistently earn abnormal returns by exploiting any information.

There are three forms of the EMH:

- Weak form: the current price of an asset reflects all past publicly available information,
- Semi-strong form: the current price of an asset reflects all publicly available information,
- Strong form: the current price of an asset reflects all information, both public and private.

EMH and The General Dynamics of Asset Returns

Under the EMH, the asset return process can be expressed as:

$$r_t = \mu_t + \varepsilon_t,$$

where:

- μ_t is the rational expectation (or the least square prediction) of r_t based on all publicly available information at time $t-1$; according to the stylized fact, we may set $\mu_t = \mu$ as a constant;
- ε_t is the unpredictable return termed innovation due to new information arriving between $t-1$ and t, specified as a white noise process:
  - ε_t is serially uncorrelated, i.e., $\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$ for $k \geq 1$,
  - $\text{Var}(\varepsilon_t) = \sigma^2$.

Two Stronger Assumptions on the Innovations

IID innovations:

$$\varepsilon_t \sim \text{i.i.d.}(0, \sigma^2).$$

Martingale difference (MD) innovations:

$$\mathbb{E}[\varepsilon_t | r_{t-1}, r_{t-2}, \cdots] = \mathbb{E}[\varepsilon_t | \varepsilon_{t-1}, \varepsilon_{t-2}, \cdots] = 0,$$

where $\varepsilon_{t-1}, \varepsilon_{t-2}, \cdots$ are the innovations at time $t-1, t-2, \cdots$.

A frequently used example of martingale difference innovations:

$$\varepsilon_t = \sigma_t \eta_t,$$

where $\eta_t \sim \text{i.i.d.}(0, 1)$.

The Relationship Between the Three Types of Innovations

$$\text{IID} \implies \text{Martingale Difference} \implies \text{White Noise.}$$

The i.i.d. innovations are the strongest assumption but it contradicts the stylized fact that the squared returns are long-range dependent, while the white noise assumption is too general to operate (the violation of zero series correlation may lead to the violation of EMH).

The martingale difference innovations are a compromise between the i.i.d. innovations and the white noise assumption, and it is the most commonly used assumption.

- The returns are not predictable;
- but volatility may be predictable.

Tests for White Noise: the Ljung-Box Test

The Ljung-Box $Q(m)$ statistic is defined as:

$$Q(m) = T(T+2)\sum_{k=1}^{m}\frac{\hat{\rho}(k)}{T-k},$$

Sample ACF

where $\hat{\rho}(k)$ is the sample autocorrelation function (ACF) of the returns and m is a prespecified positive integer.

We reject the null hypothesis of white noise at the significance level α if $Q(m) > \chi^2_{1-\alpha}(m)$, where $\chi^2_{1-\alpha}(m)$ is the $(1-\alpha)$ quantile of the chi-square distribution with m degrees of freedom or the p-value $\mathbb{P}(Q > Q(m)) < \alpha$ with $Q \sim \chi^2(m)$.

Ljung-Box Test for S&P 500 Returns

Number of returns data points: 502, from 2023-01-01 to 2024-12-31.

Ljung-Box Test Results			
	$m = 6$	$m = 12$	$m = 24$
Returns			
Q-stat	4.5393	7.6178	24.4364
p-value	0.6041	0.8142	0.4369
Absolute Returns			
Q-stat	16.6896	28.7464	35.9555
p-value	0.0105	0.0043	0.0554
Squared Returns			
Q-stat	19.7053	32.8507	36.6592
p-value	0.0031	0.0010	0.0473

Using too long data may lead to the rejection of the null hypothesis of white noise.

Augmented Dickey-Fuller (ADF) Test

The ADF test is a popular test for the null hypothesis of a unit root, i.e., the returns are non-stationary.

There are three versions of the ADF test:

- no constant/trend: $\ln S_t = \alpha \ln S_{t-1} + \sum_{i=1}^{p} \beta_i \Delta \ln S_{t-i} + \varepsilon_t$;
- constant only: $\ln S_t = \alpha \ln S_{t-1} + \beta + \sum_{i=1}^{p} \beta_i \Delta \ln S_{t-i} + \varepsilon_t$;
- constant and trend: $\ln S_t = \alpha \ln S_{t-1} + \beta + \gamma t + \sum_{i=1}^{p} \beta_i \Delta \ln S_{t-i} + \varepsilon_t$.

Null hypothesis: $\alpha = 1$; alternative hypothesis: $\alpha < 1$. Test statistic:

$$\text{DF} = \frac{\widehat{\alpha} - 1}{\text{SE}(\widehat{\alpha})}.$$

ADF Test for S&P 500 Returns

Augmented Dickey-Fuller Test Results for Log SPX Price			
	No Const/Trend	Const Only	Const& Trend
Test Statistic	1.7952	0.6539	−2.7549
p-value	0.9832	0.9889	0.2140
1% Critical Value	−2.5661	−3.4314	−3.9603
5% Critical Value	−1.9410	−2.8620	−3.4112
10% Critical Value	−1.6168	−2.5670	−3.1275

Augmented Dickey-Fuller Test Results for SPX Returns			
	No Const/Trend	Const Only	Const& Trend
Test Statistic	−21.3870	−21.5970	−21.5787
p-value	0.0000	0.0000	0.0000
1% Critical Value	−2.5702	−3.4435	−3.9770
5% Critical Value	−1.9415	−2.8673	−3.4193
10% Critical Value	−1.6163	−2.5699	−3.1322

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Stationary Time Series

A stationary[1] time series is a time series $\{X_t\}$ that has a constant mean and variance, and the covariance between the returns at different times is constant, i.e.,

$$\mathbb{E}[X_t], \quad \text{Var}(X_t), \quad \text{Cov}(X_t, X_{t-k}),$$

are independent of t.

The ACF of a stationary time series is given by:

$$\rho(k) = \frac{\gamma_k}{\gamma_0},$$

where $\gamma_k = \text{Cov}(X_t, X_{t-k})$.

[1] Strong stationary means the joint distribution of $(X_t, X_{t-1}, \cdots, X_{t-k})$ is the same as that of $(X_{t+h}, X_{t+h-1}, \cdots, X_{t+h-k})$ for any $h \geq 0$ and t

Moving Average (MA) Models

The MA(q) model is defined as:

$$X_t = \mu + \varepsilon_t + \sum_{i=1}^{q} a_i \varepsilon_{t-i},$$

where $\varepsilon_t \sim \text{WN}(0, \sigma^2)$ is the unobservable innovations at time t, and $a_1, a_2, \cdots, a_q$ are the parameters.

The MA(q) model is always stationary.

$$\mathbb{E}[X_t] = \mu,$$

$$\text{Var}(X_t) = \sigma^2(1 + a_1^2 + a_2^2 + \cdots + a_q^2),$$

$$\text{Cov}(X_t, X_{t-|k|}) = \sigma^2 \left(a_{|k|} + \sum_{i=1}^{q-|k|} a_{|k|+i} a_i \right), \quad |k| \le q,$$

$$\text{Cov}(X_t, X_{t-k}) = 0, \quad |k| > q.$$

$$X_t = \mu + \varepsilon_t + a_1 \varepsilon_{t-1} + a_2 \varepsilon_{t-2} + a_k \varepsilon_{t-k} + a_q \varepsilon_{t-q}$$

$$X_{t-k} = \mu + \varepsilon_{t-k} + a_1 \varepsilon_{t-k-1} + \cdots + a_{q-k} \varepsilon_{t-q} + \cdots$$

$$a_k \cdot \sigma^2 + a_1 a_{k+1} \cdot \sigma^2 + a_2 a_{k+2} \sigma^2 + \cdots + a_q \cdot a_{q-k} \sigma^2$$

The ACF of the MA(q) Model

The ACF of the MA(q) model is given by:

$$\rho(k) = \begin{cases} \dfrac{a_{|k|} + \sum_{i=1}^{q-|k|} a_{|k|+i} a_i}{1 + a_1^2 + a_2^2 + \cdots + a_q^2}, & |k| \le q, \\ 0, & |k| > q. \end{cases}$$

The ACF of the MA(q) model is independent of σ^2, μ, and cuts off after q lags.

To Determine the Order of the MA Model

It is known that the ACF of the MA(q) model cuts off after q lags. Further assuming that $\varepsilon_t \sim$ i.i.d.$(0, \sigma^2)$ and $\mathbb{E}[\varepsilon_t^4] < \infty$, we have that

$$\sqrt{T}\widehat{\rho}(k) \xrightarrow{d} \mathcal{N}\left(0, 1 + 2\sum_{i=1}^{q} \rho(i)^2\right), \quad k > q,$$

as the sample size T goes to infinity, where $\widehat{\rho}(k)$ is the sample ACF of the returns.

We can use the above equation to determine the order of the MA model by testing the null hypothesis that $\widehat{\rho}(k) = 0$ and set q as the largest integer such that the null hypothesis is rejected.

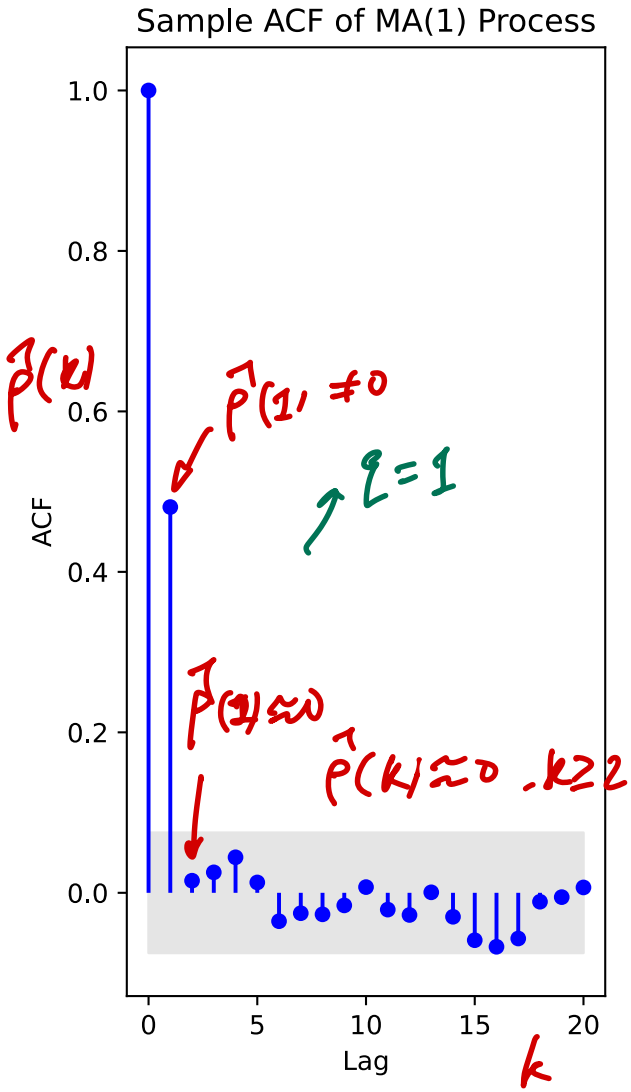
To Determine the Order of the MA Model (continued)

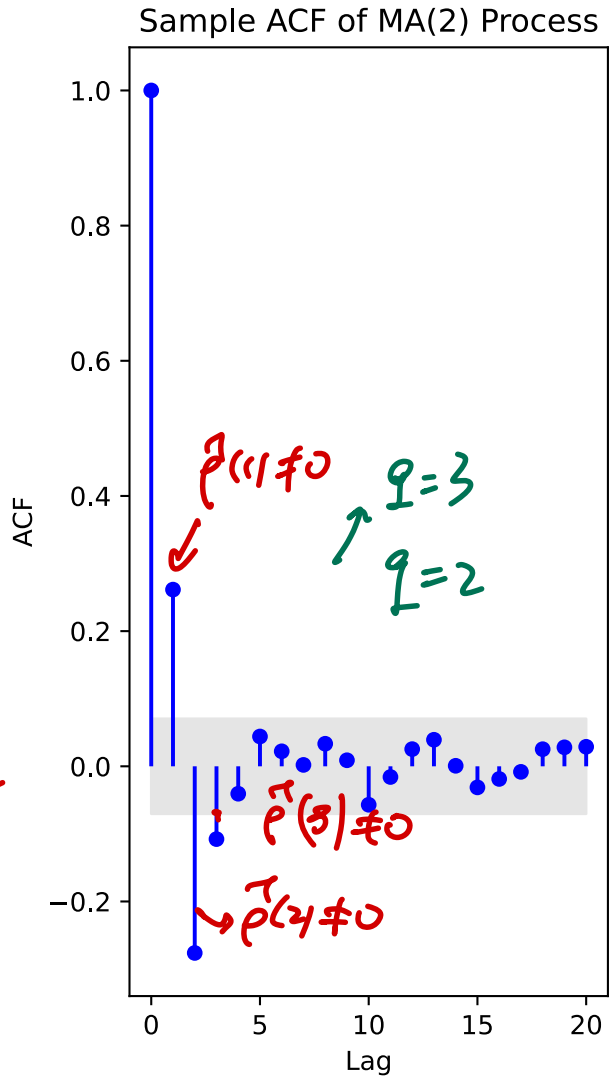
As an example, we generate 1000 observations from the following models.

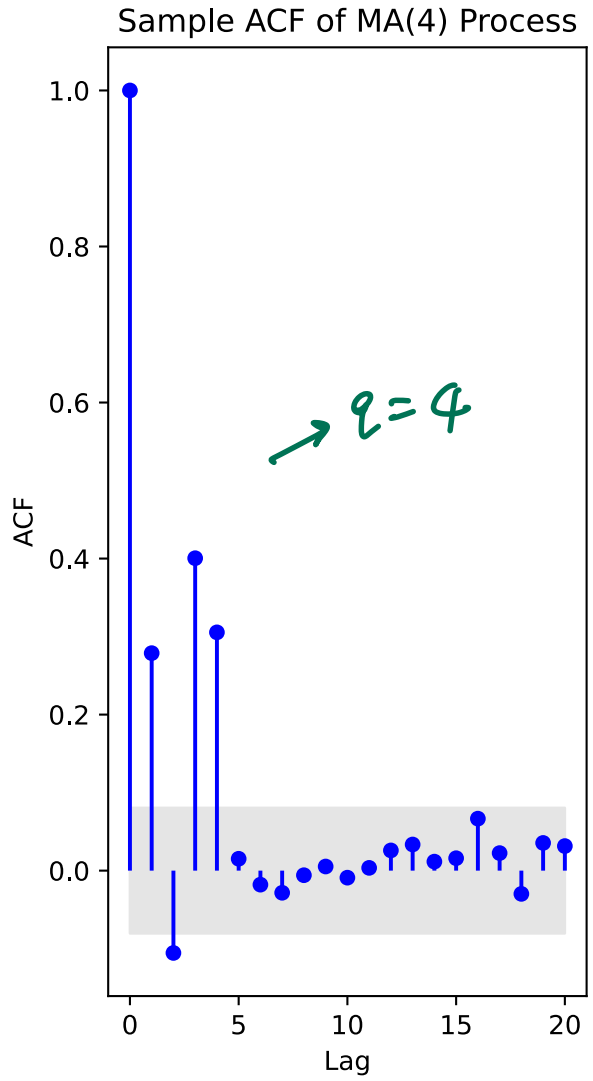
- $MA(1)$ model with $a_1 = 0.7$,
- $MA(2)$ model with $a_1 = 0.7$ and $a_2 = -0.4$,
- $MA(4)$ model with $a_1 = 0.7$, $a_2 = -0.4$, $a_3 = 0.6$, and $a_4 = 0.8$.

We plot the sample ACF and the 95% confidence intervals.

To Determine the Order of the MA Model (continued)







Identification and Invertibility of MA Models

Consider two MA(1) models:

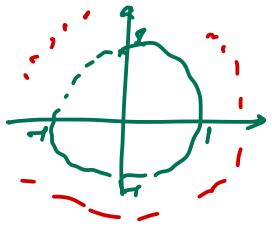
$$X_t = \varepsilon_t + a\varepsilon_{t-1}, \ \varepsilon_t \sim \text{i.i.d.}(0, \sigma^2),$$

$$Y_t = e_t + a^{-1}e_{t-1}, \ e_t \sim \text{i.i.d.}(0, a^2\sigma^2).$$

The two models are observationally equivalent, i.e., they have the same ACVF and hence the same ACF. Therefore, we cannot identify the two models from the ACF. To solve this issue, we can impose that $|a| < 1$.

This is equivalent to the invertibility condition of the MA model. For MA(q) models, the invertibility condition is that the roots of the characteristic polynomial $1 + a_1 z + a_2 z^2 + \cdots + a_q z^q = 0$ are all outside the unit circle.

$$X_t = \varepsilon_t + a\varepsilon_{t-1}$$



MA(1)

$1+a_1 z = 0$

$z = -\frac{1}{a_1}$

$\left|\frac{1}{a_1}\right| > 1$

$$\varepsilon_t = X_t - a\varepsilon_{t-1}$$

$$= X_t - a\cdot\left[X_{t-1} - a\varepsilon_{t-2}\right] = X_t - aX_{t-1} + a^2\varepsilon_{t-2}$$

$$= X_t - aX_{t-1} + a^2\cdot\left[X_{t-2} - a\varepsilon_{t-3}\right]$$

$$= X_t + \sum_{i=1}^{N} (-a)^i X_{t-i} + (-a)^{N+1}\varepsilon_{t-N-1}$$

$N \to \infty$ $\quad (-a)^{N+1} \to 0$

$|a| < 1$

Autoregressive (AR) Models

The AR(p) model is defined as:

$$X_t = c + \sum_{i=1}^{p} b_i X_{t-i} + \varepsilon_t,$$

where $\varepsilon_t \sim \text{WN}(0, \sigma^2)$ is the unobservable innovations at time t, and $b_1, b_2, \cdots, b_p$ are the parameters.

The AR(p) model is not always stationary. For example, the AR(1) model $X_t = bX_{t-1} + \varepsilon_t$ is stationary if $|b| < 1$, and non-stationary if $|b| \geq 1$.

Stationary AR(p) Models

The AR(p) model is stationary if the p roots of the characteristic polynomial $1 - b_1 z - b_2 z^2 - \cdots - b_p z^p = 0$ are all outside the unit circle, i.e., the modulus of the roots are greater than 1.

The ACF of the stationary AR(p) model decays exponentially, i.e., $\rho(k) = \mathcal{O}(a^k)$ for some $a < 1$ as $k \to \infty$.

Suppose the AR(p) model is stationary, the mean is given by:

$$\mathbb{E}[X_t] = \frac{c}{1 - b_1 - b_2 - \cdots - b_p}.$$

$X_t = c + \sum_{i=1}^{p} b_i X_{t-i} + \varepsilon_t$

$E[X_t] = c + \sum_{i=1}^{p} b_i E(X_{t-i}) + E[\varepsilon_t] = c + E[X_t] \sum_{i=1}^{p} b_i$

$E[X_t]$ $\quad -0$

Stationary AR(p) Models (continued)

The ACVF satisfies the so-called Yule-Walker equations:

$$\gamma(k) = b_1\gamma(k-1) + b_2\gamma(k-2) + \cdots + b_p\gamma(k-p), \quad k \geq 1.$$

$Cov(X_t, X_{t-k})$

We can first solve the above equations for $\gamma(k)$ for $k \leq p$, and then use the Yule-Walker equations to calculate the ACVF for $k > p$.

For example, consider the AR(1) model $X_t = bX_{t-1} + \varepsilon_t$. The ACVF and ACF are given by:

$\gamma(0) = b\gamma(1) + \sigma^2$

$\gamma(1) = b\gamma(0)$

$\gamma(0) = b^2\gamma(0) + \sigma^2$

$\gamma(0) = \frac{\sigma^2}{1-b^2}$

$\gamma(1) = b\sigma^2/(1-b^2)$

$$\gamma(k) = \begin{cases} \sigma^2/(1-b^2), & k = 0, \\ \sigma^2 b^{|k|}/(1-b^2), & |k| > 0. \end{cases}$$

$$\rho(k) = \begin{cases} 1, & k = 0, \\ b^{|k|}, & |k| > 0. \end{cases}$$

$k \geq 1$:

$\gamma(k) = b\gamma(k-1)$

$= b^k \cdot \gamma(0)$

$= b^k \cdot \frac{\sigma^2}{1-b^2}$

$\gamma(k) = Cov(X_t, X_{t-k})$ $\quad k \geq 0$

$= Cov(c + \sum_{i=1}^{p} b_i X_{t-i} + \varepsilon_t, \; X_{t-k})$

$= \sum_{i=1}^{p} b_i \cdot Cov(X_{t-i}, X_{t-k}) + Cov(\varepsilon_t, X_{t-k})$

$Cov(X_{t-i}, X_{t-k}) = \gamma(i-k)$

$Cov(\varepsilon_t, X_{t-k}) = \sigma^2 \cdot \mathbb{1}_{\{k=0\}}$

$\varepsilon_{t-1}, \varepsilon_{t-2} \cdots$

$c + \sum_{i=1}^{p} b_i X_{t-k-i} + \varepsilon_{t-k}$

$t-k-i \leq t-1$

$\gamma(0) = b_1 \gamma(1) + b_2 \gamma(2) + \cdots + b_p \gamma(p) + \sigma^2$

$\gamma(1) = b_1 \gamma(0) + b_2 \gamma(1) + \cdots + b_p \gamma(p-1)$

$\gamma(2) = b_1 \gamma(1) + b_2 \gamma(0) + \cdots + b_p \gamma(p-2)$

$\gamma(3) = b_1 \gamma(2) + b_2 \gamma(1) + \cdots + b_p \gamma(p-3)$

$\gamma(p) = b_1 \gamma(p-1) + b_2 \gamma(p-2) + \cdots + b_p \gamma(0)$

$\gamma(k) = \gamma(-k)$

$Cov(X_t, X_{t+c}) \quad Cov(X_{t+c}, X_t)$

Autoregressive (AR) Models are MA(∞) Models

Consider the AR(1) model $X_t = bX_{t-1} + \varepsilon_t$. We can write the model as:

$$X_t = \varepsilon_t + b\varepsilon_{t-1} + b^2\varepsilon_{t-2} + \cdots$$
$$= \sum_{i=0}^{\infty} b^i \varepsilon_{t-i},$$

where the convergence is in mean square when $|b| < 1$.

The above derivation can be reformulated using the lag operator B, i.e., $BX_t = X_{t-1}$. Then we have:

$$X_t = (1 - bB)^{-1}\varepsilon_t = \sum_{i=0}^{\infty} b^i B^i \varepsilon_t = \sum_{i=0}^{\infty} b^i \varepsilon_{t-i},$$

where the convergence is in mean square when $|b| < 1$.

Autoregressive models are always invertible.

AR Models are Mean Reverting

Let

$$\mu = \mathbb{E}[X_t] = \frac{c}{1 - \sum_{i=1}^{p} b_i}.$$

Then,

$$X_t - \mu = \sum_{i=1}^{p} b_i (X_{t-i} - \mu) + \varepsilon_t.$$

To Determine the Order of the AR Model

The partial autocorrelation function (PACF) is defined as the correlation between X_{k+1} and X_1 after removing the linear dependence on the intermediate variables $X_2, X_3, \cdots, X_k$.

Suppose we run a regression:

$$\min_{\beta_0,\beta_1,\beta_2,\cdots,\beta_k} \mathbb{E}\left[\left(X_{k+1} - \beta_0 - \sum_{i=1}^{k} \beta_i X_{k-i+1}\right)^2\right].$$

The PACF is given by $\pi(k) = \beta_k^*$, which cuts off after p lags, i.e., $\pi(k) = 0$ for $k > p$.

To Determine the Order of the AR Model (continued)

The sample PACF can be estimated by a regression based on the observed series:

$$\min_{\beta_0,\beta_1,\beta_2,\cdots,\beta_k} \sum_{t=k+1}^{T} \left(X_t - \beta_0 - \sum_{i=1}^{k} \beta_i X_{t-i} \right)^2 .$$

Suppose the true model is an stationary AR(p) model with $\varepsilon_t \sim$ i.i.d.$(0, \sigma^2)$ and $\mathbb{E}[\varepsilon_t^4] < \infty$,

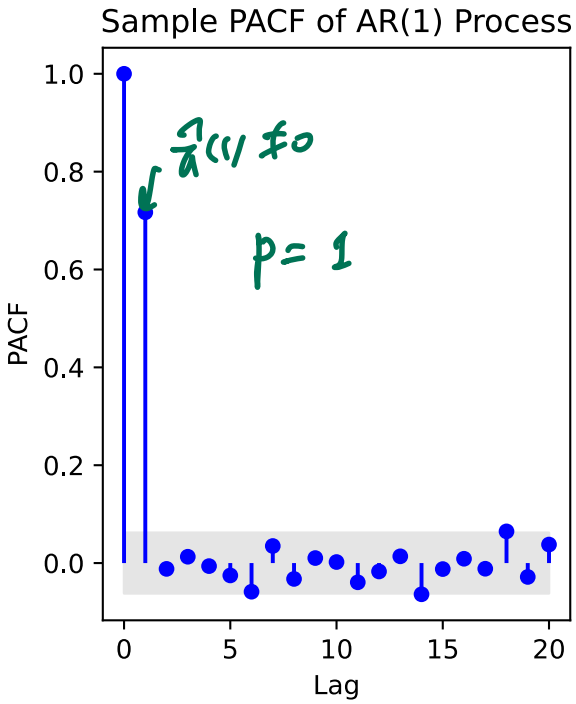
$$\sqrt{T}\widehat{\pi}(k) \xrightarrow{d} \mathcal{N}(0,1), \quad k > p.$$

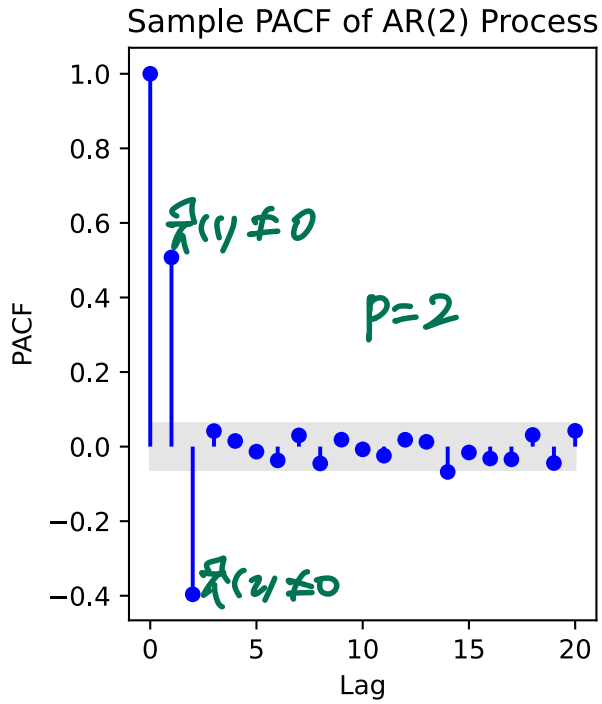
To Determine the Order of the AR Model (continued)

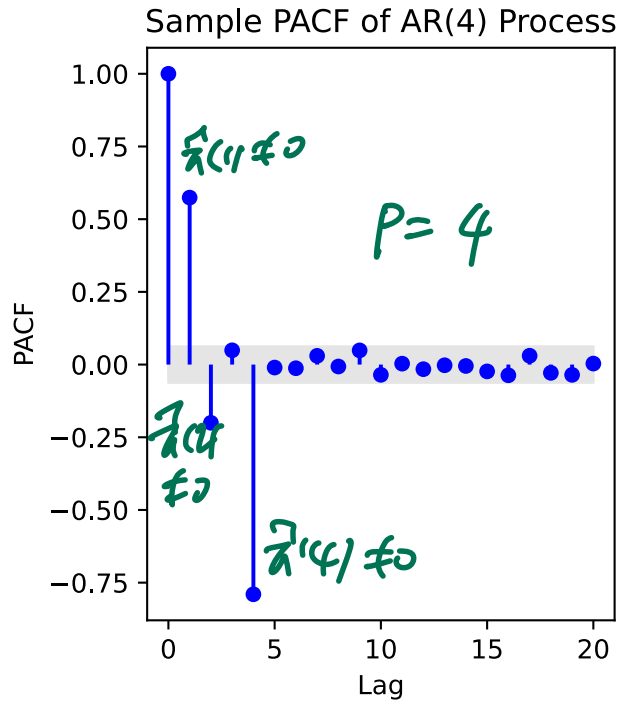
As an example, we generate 1000 observations from the following models.

- $AR(1)$ model with $b_1 = 0.7$,
- $AR(2)$ model with $b_1 = 0.7$ and $b_2 = -0.4$,
- $AR(4)$ model with $b_1 = 0.7$, $b_2 = -0.4$, $b_3 = 0.6$, and $b_4 = -0.8$.

To Determine the Order of the AR Model (continued)







ARMA Models

$X_t = \mu_t + \varepsilon_t$

$\mu_t = c + \sum_{i=1}^{p} b_i X_{t-i} + \sum_{i=1}^{q} a_i \varepsilon_{t-i}$

The ARMA(p, q) model is defined as:

$$X_t = c + \sum_{i=1}^{p} b_i X_{t-i} + \varepsilon_t + \sum_{i=1}^{q} a_i \varepsilon_{t-i},$$

AR MA

where $\varepsilon_t \sim \text{WN}(0, \sigma^2)$ is the unobservable innovations at time t, and $b_1, b_2, \cdots, b_p$ and $a_1, a_2, \cdots, a_q$ are the parameters.

The stationary condition of the ARMA(p, q) model is that the p roots of the characteristic polynomial $b(z) = 1 - b_1 z - b_2 z^2 - \cdots - b_p z^p$ are all outside the unit circle.

Same as the AR(p) model

The invertible condition of the ARMA(p, q) model is that the q roots of the characteristic polynomial $a(z) = 1 + a_1 z + a_2 z^2 + \cdots + a_q z^q$ are all outside the unit circle.

Same as MA(q)

Stationary ARMA Models

For a stationary ARMA(p, q) model, the mean, ~~variance, and autocovariance function are~~ given by:

$$\mathbb{E}[X_t] = \mu = \frac{c}{1 - b_1 - b_2 - \cdots - b_p}.$$

The ARMA model can be written as:

$$b(B)X_t = c + a(B)\varepsilon_t.$$

If it is stationary, we can write the ARMA model as:

$$X_t = \mu + a(B)^{-1}(1 + b(B))\varepsilon_t = \mu + \varepsilon_t + \sum_{i=1}^{\infty} \alpha_i \varepsilon_{t-i},$$

with $\sum_{i=1}^{\infty} |\alpha_i| < \infty$, which is an MA$(\infty)$ process (belongs to the class of causal processes).

Invertibility of the ARMA Model

For an invertible ARMA(p, q) model, we can write the ARMA model as:

$$\begin{aligned}\varepsilon_t &= a^{-1}(B)\left(b(B)X_t - c\right) \\ &= X_t - \mu^* - \sum_{i=1}^{\infty} \beta_i X_{t-i},\end{aligned}$$

for some constants μ^* and β_i. This is an AR(∞) process.

For example, consider the ARMA$(1,1)$ model $X_t = c + bX_{t-1} + \varepsilon_t + a\varepsilon_{t-1}$. We can write the model as:

$$\begin{aligned}\varepsilon_t &= (1+aB)^{-1}\left(X_t - bX_{t-1} - c\right) \\ &= (1 - aB + a^2B^2 - \cdots)\left(X_t - bX_{t-1} - c\right) \\ &= \sum_{i=0}^{\infty} (-a)^i B^i \left(X_t - bX_{t-1} - c\right).\end{aligned}$$

The Autocorrelation Functions of the ARMA Model

Autocorrelation:

- The autocorrelation function of an ARMA(p, q) process exhibits exponential decay towards zero: it does not cut off but gradually dies out as $|k|$ increases, possibly damped oscillations.
- The autocorrelation function of an ARMA(p, q) process displays the shape of that of an AR(p) process for $|k| > \max(p, q + 1)$.

Partial Autocorrelation:

- The partial autocorrelation function of an ARMA(p, q) process will gradually die out (the same property as a MA(q) model).

It is more difficult to determine the order of the ARMA model than the AR and MA models.

Gaussian MLE of ARMA Models

Though the innovations are not necessarily Gaussian, we can still use the Gaussian maximum likelihood estimation (MLE) to estimate the parameters of the ARMA model which can lead to consistent estimates of the parameters.

In this case, $\boldsymbol{X} = (X_1, X_2, \cdots, X_T)$ are jointly normal with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ depending on the model parameters $\boldsymbol{\theta} = (c, \boldsymbol{a}, \boldsymbol{b}, \sigma^2)$. Then joint density function of $\boldsymbol{X}$ is given by:

$$f(\boldsymbol{X}) = \frac{1}{(2\pi)^{T/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\boldsymbol{X}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{X}-\boldsymbol{\mu})\right).$$

The log-likelihood function is given by:

$$\ell(\boldsymbol{\theta}) = \log L(\boldsymbol{\theta}) = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\log|\boldsymbol{\Sigma}| - \frac{1}{2}(\boldsymbol{X}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{X}-\boldsymbol{\mu}).$$

If the true distribution is not Gaussian, the resulting estimators are quasi-MLE.

Statistical Inference of ARMA Models

Once we have the MLE $\widehat{\boldsymbol{\theta}}$ of the parameters, we can conduct the statistical inference of the parameters.

For example, we can conduct the Wald test to test the significance of the parameters.The Wald test is given by:

$$W = (\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)^T \widehat{\text{Var}}(\widehat{\boldsymbol{\theta}})^{-1} (\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \overset{a}{\sim}_{H_0} \chi^2(p),$$

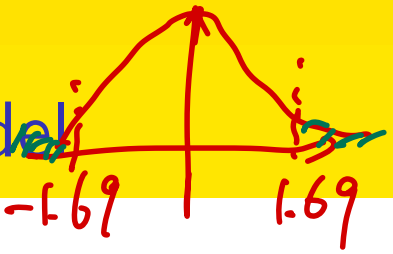
where p is the number of parameters, $\widehat{\text{Var}}(\widehat{\boldsymbol{\theta}}) = -\partial^2 \ell(\widehat{\boldsymbol{\theta}}) / \partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^T$ is the covariance matrix of $\widehat{\boldsymbol{\theta}}$ (also called the Fisher information matrix).

Suppose we want to calculate the standard error of $f(\widehat{\boldsymbol{\theta}})$ for some smooth function f, we can use the Delta Method:

$$\text{Var}(f(\widehat{\boldsymbol{\theta}})) \approx \partial f(\widehat{\boldsymbol{\theta}}) / \partial \boldsymbol{\theta}^T \widehat{\text{Var}}(\widehat{\boldsymbol{\theta}}) \partial f(\widehat{\boldsymbol{\theta}}) / \partial \boldsymbol{\theta}.$$

Aympotically, we can assume that the distribution of $\widehat{\boldsymbol{\theta}}$ is Gaussian.

Fitting SPX Returns to an ARMA(1, 1) Model



We fit the SPX returns to an ARMA(1, 1) model and report the summary of the model.

Parameter	Coef	Std Err	Z	p value
const	0.0002	0.000	1.690	0.091
ar.L1	−0.0129	0.048	−0.266	0.790
ma.L1	−0.0922	0.049	−1.869	0.062
sigma2	0.0001	1.12×10^{-6}	130.141	0.000

For the unconditional mean, we use the Delta method to calculate the standard error and the 95% confidence interval:

Metric	Value
Estimated Mean	2.46×10^{-4}
Standard Error	1.44×10^{-4}
95% CI	$[-3.60, 52.9] \times 10^{-5}$

To Determine the Order of the ARMA Model

For ARMA models, there is no simple way to determine the order of the model.

We can use the AIC (Akaike Information Criterion) or BIC (Bayesian Information Criterion) to determine the order of the model.

The AIC and BIC are defined as:

$$\text{AIC} = -2\log\widehat{L} + 2(p+q+1),$$

$$\text{BIC} = -2\log\widehat{L} + \log(T)(p+q+1),$$

where $\widehat{L}$ is the maximized log-likelihood function and p and q are the orders of the AR and MA parts, respectively.

Fitting SPX Returns to ARMA Models

We fit the SPX returns to ARMA models with different orders (p and q) and report the AIC and BIC of the models.

p	q	AIC	BIC
0	0	−36103.89	−36090.47
0	1	−36168.02	−36147.91
0	2	−36166.03	−36139.21
0	3	−36164.12	−36130.60
0	4	−36169.50	−36129.27
1	0	−36167.44	−36147.32
1	1	−36166.04	−36139.22
1	2	−36164.03	−36130.50
1	3	−36163.01	−36122.78
1	4	−36166.37	−36119.44
2	0	−36166.15	−36139.34
2	1	−36163.94	−36130.41

The Best AIC and BIC Models

Better to specify a MA(4) or MA(4) for SPX returns

Best AIC Model Summary:

	coef	std err	z	$P > \|z\|$
const	0.0002	0.000	1.652	0.099
ma.L1	-0.1038	0.007	-15.492	0.000
ma.L2	-0.0010	0.005	-0.192	0.848
ma.L3	0.0123	0.007	1.885	0.059
ma.L4	-0.0314	0.006	-5.202	0.000
sigma2	0.0001	1.14e-06	127.352	0.000

Best BIC Model Summary:

	coef	std err	z	$P > \|z\|$
const	0.0002	0.000	1.724	0.085
ma.L1	-0.1051	0.006	-16.204	0.000
sigma2	0.0001	1.11e-06	130.871	0.000

Model Diagnostics: Residual Analysis

After fitting a model, we can calculate the residuals:

$$\widehat{\varepsilon}_t = X_t - \sum_{i=1}^{p} \widehat{b}_i X_{t-i} - \sum_{i=1}^{q} \widehat{a}_i \widehat{\varepsilon}_{t-i} - \widehat{c}, \quad t = p+1, p+2, \cdots, T,$$

where the initial values $\widehat{\varepsilon}_{p+1-i}, i = 1, 2, \cdots, q$ can be set to zero.

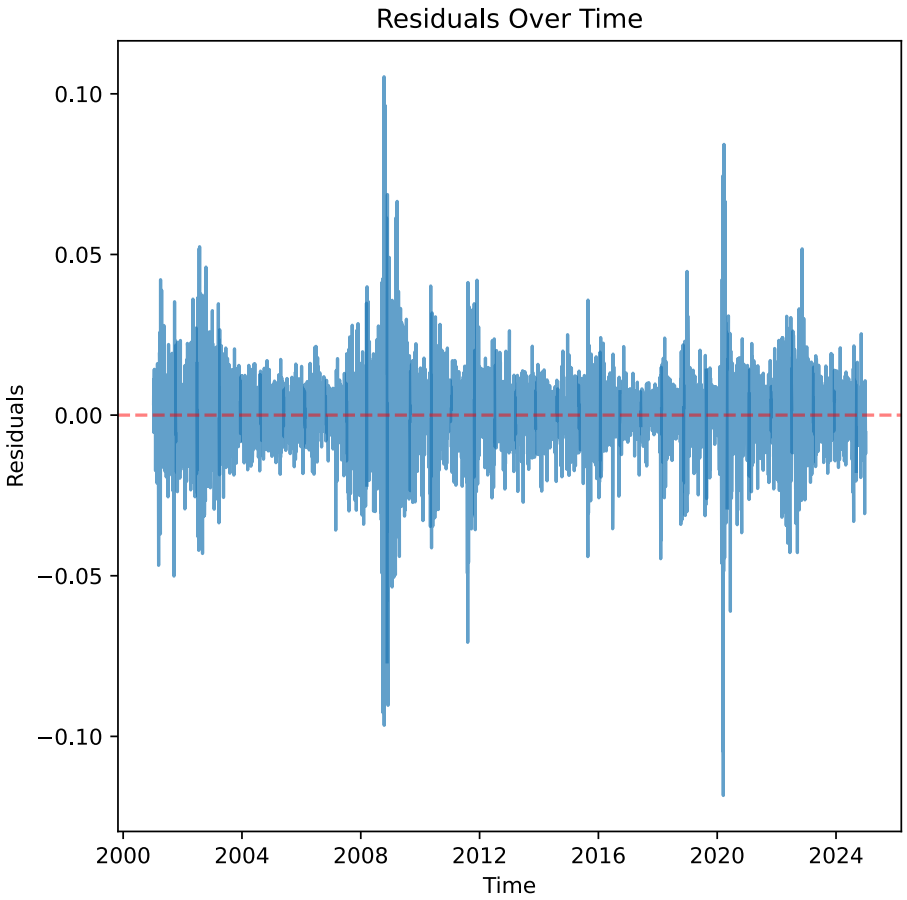
Most computer packages revert to AR(∞) to calculate the residuals.

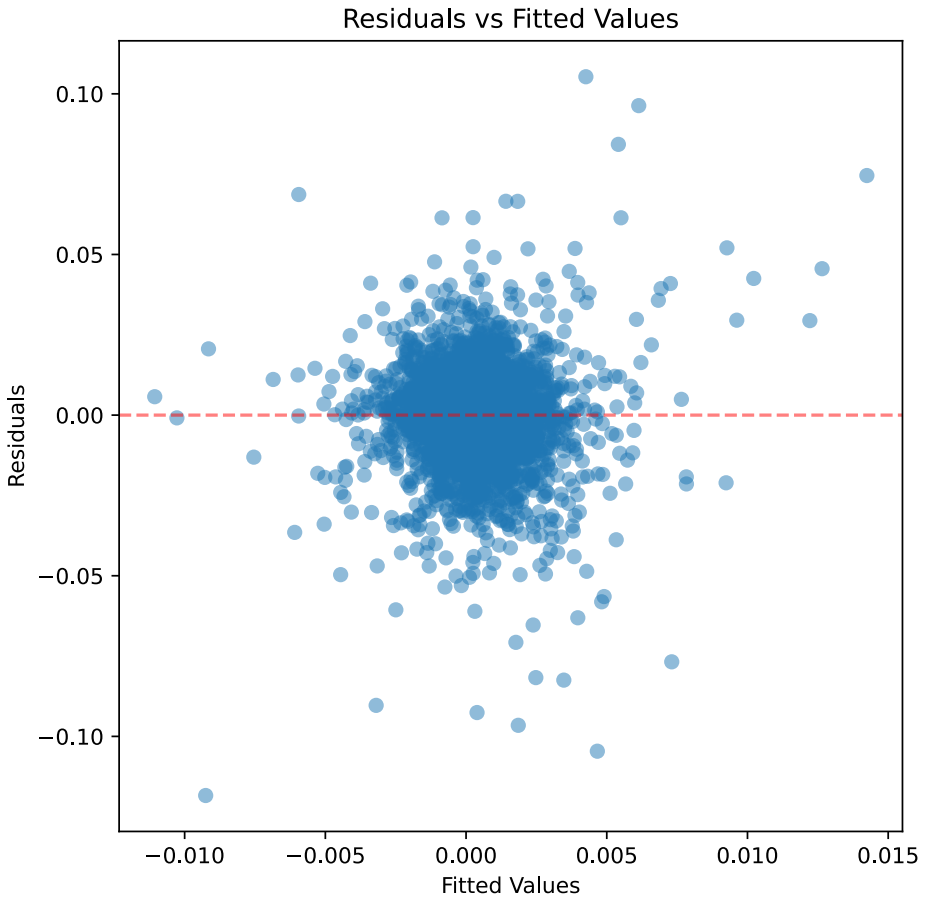
Residual plots are useful for checking the model adequacy.

- Plot residuals against time or fitted values.
- Plot the autocorrelation function of the residuals.
- Plot the QQ plot of the residuals.

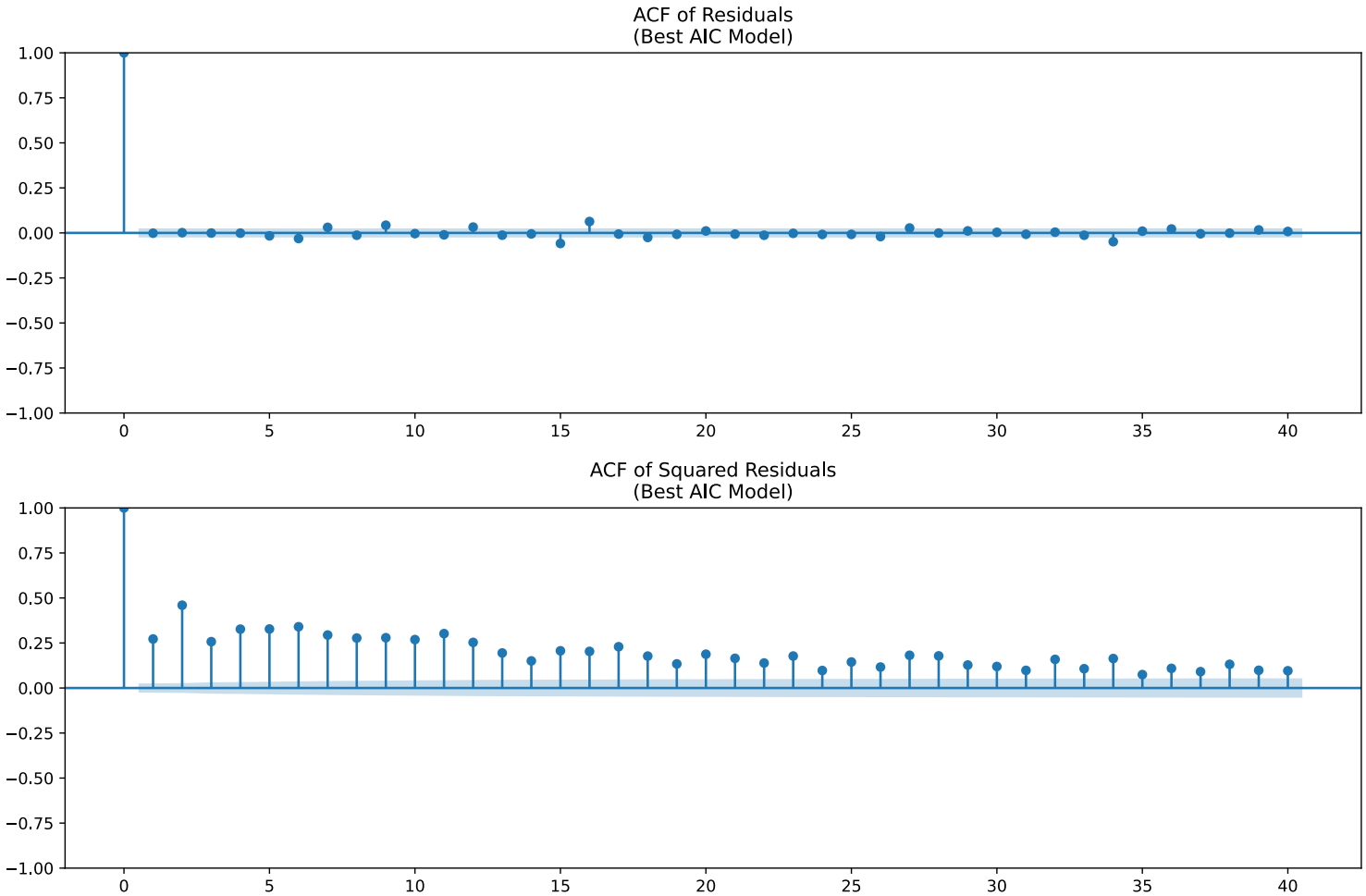
Also, we can use the Ljung-Box test to test the significance of the autocorrelations of the residuals, with the degrees of freedom $m - p - q$, where m is the lag length.

Residual Plots: Best AIC Model





Residual ACF Plots: Best AIC Model



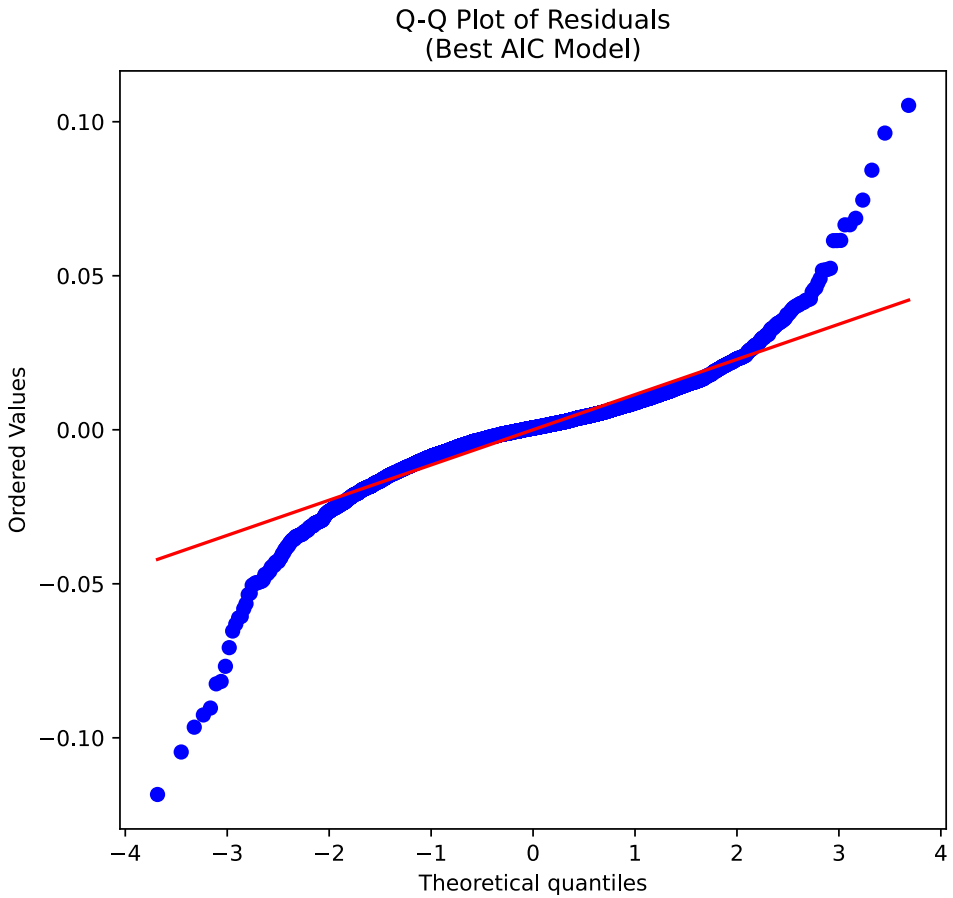
Ljung-Box Test: Best AIC Model

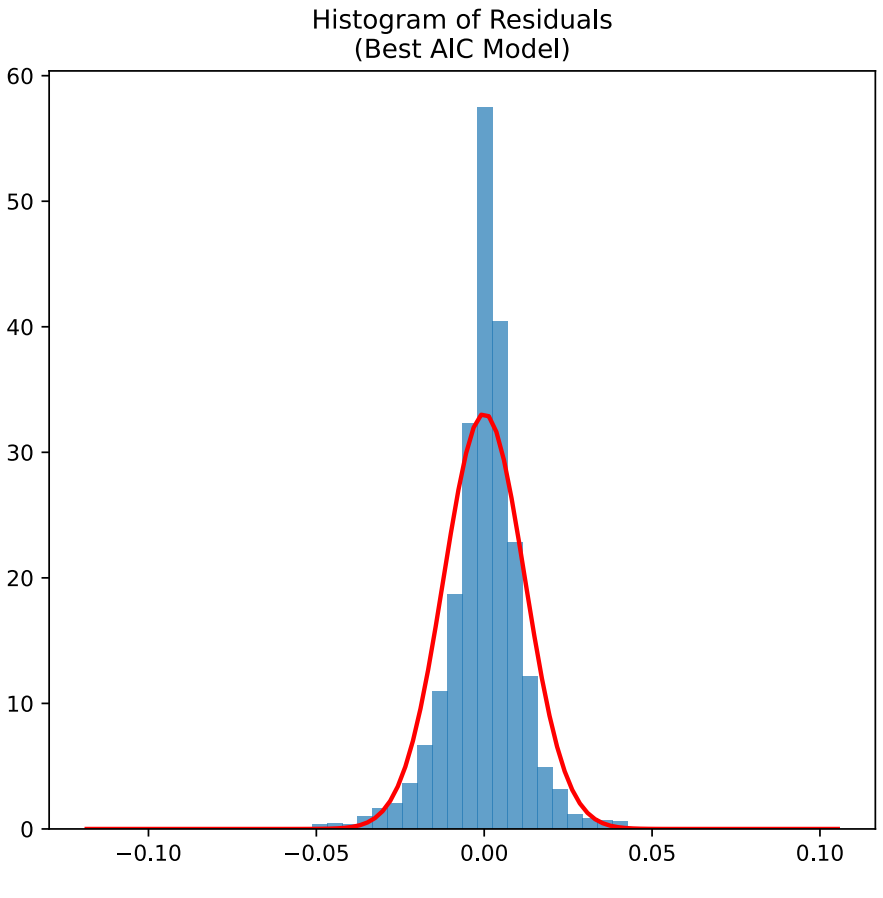
Lag	Statistic	p-value
	$\widehat{\varepsilon}_t$	
10	25.46	0.0001
15	54.39	0.0000
20	83.70	0.0000
	$\lvert\widehat{\varepsilon}_t\rvert$	
10	6638.29	0.0000
15	8873.29	0.0000
20	10656.59	0.0000
	$\lvert\widehat{\varepsilon}_t\rvert^2$	
10	6010.40	0.0000
15	7573.39	0.0000
20	8653.20	0.0000

Residual QQ Plots: Best AIC Model

Jarque-Bera Test Results:

- Statistic: 26788.88
- p-value: 0.0000





Forecasting with ARMA Models

In general, we are interested in the point prediction of the future values of the series, given that the model parameters have been estimated.

$$X_{T+h} = c + \sum_{i=1}^{p} b_i X_{T+h-i} + \varepsilon_{T+h} + \sum_{i=1}^{q} a_i \varepsilon_{T+h-i}.$$

Taking expectation on both sides based on the information up to time T, we get:

$$\mathbb{E}_T[X_{T+h}] = c + \sum_{i=1}^{p} b_i \mathbb{E}_T[X_{T+h-i}] + \sum_{i=1}^{q} a_i \mathbb{E}_T[\varepsilon_{T+h-i}],$$

where

$$\mathbb{E}_T[\varepsilon_{T+h-i}] = \begin{cases} 0, & i < h, \\ \varepsilon_{T+h-i}, & i \geq h. \end{cases}$$

Forecasting with ARMA Models (continued)

To calculate the prediction error, we reply the MA(∞) representation of the ARMA model:

$$X_t = \mu + \varepsilon_t + \sum_{i=1}^{\infty} \psi_i \varepsilon_{t-i}.$$

This implies that

$$\mathrm{Var}_T[\varepsilon_{T+j}] = \begin{cases} 0, & j \le 0 \\ E_T[\varepsilon_{T+j}^2] = E_T\left[E_{T+j-1}[\varepsilon_{T+j}^2]\right] = \sigma^2, & j \ge 1 \end{cases}$$

$$X_{T+h} = \mu + \varepsilon_{T+h} + \sum_{i=1}^{\infty} \psi_i \varepsilon_{T+h-i}.$$

Then,

$$\mathrm{Var}_T[X_{T+h}] = \mathrm{Var}_T[\varepsilon_{T+h}] + \sum_{i=1}^{\infty} \psi_i^2 \mathrm{Var}_T[\varepsilon_{T+h-i}]$$

$$\mathrm{Var}_T[X_{T+h}] = \sigma^2 \left(1 + \sum_{i=1}^{h-1} \psi_i^2\right). \qquad \begin{cases} \sigma^2 & \text{if } i \le h-1 \\ 0 & \text{if } i \ge h \end{cases}$$

ARIMA Models

Sometimes, the series is not stationary, but the first difference of the series is stationary. In this case, we can take the first difference of the series and then fit an ARMA model to the differenced series.

The ARIMA(p, d, q) model is defined as:

$$\nabla^d X_t = c + \sum_{i=1}^{p} b_i \nabla^d X_{t-i} + \varepsilon_t + \sum_{i=1}^{q} a_i \varepsilon_{t-i},$$

where ∇^d is the d-th order difference operator, i.e., $\nabla^d X_t = (\nabla^{d-1} X_t) - (\nabla^{d-1} X_{t-1})$ with $\nabla^0 X_t = X_t$.

The d-th order difference removes the trend of the series in polynomial form t^d.

Seasonal ARIMA Models

Some series are seasonal, i.e., the series has a periodic pattern. In this case, we can take the seasonal difference of the series and then fit an ARIMA model to the seasonal differenced series.

For example, the ~~monthly~~ quarterly sales of iPhones are seasonal, and the seasonal difference is taken to remove the seasonal pattern:

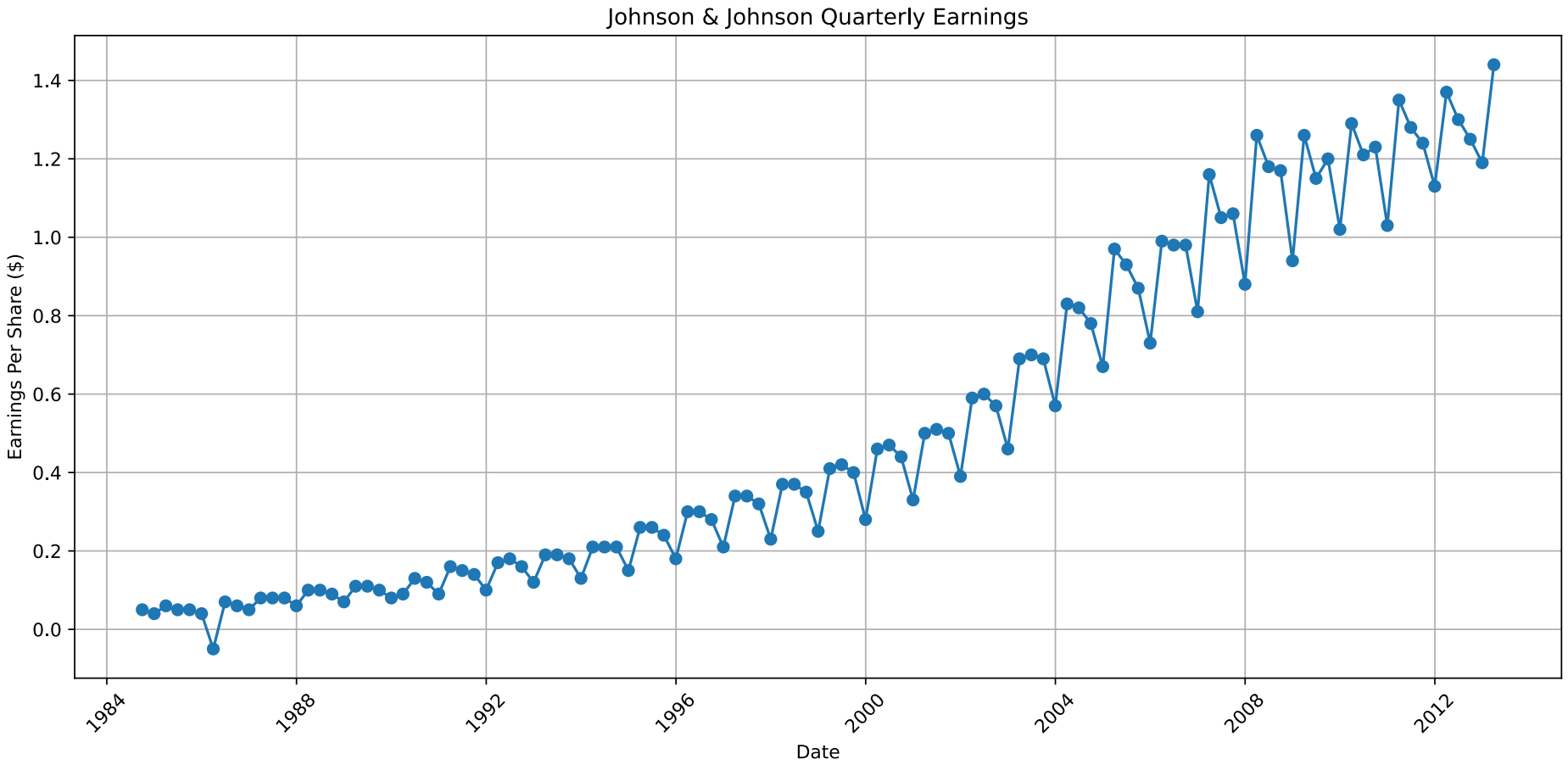
$$Y_t = X_t - X_{t-4} = (1 - B^4)X_t.$$

$BX_t = X_{t-1}$

$B^2X_t = X_{t-2}$

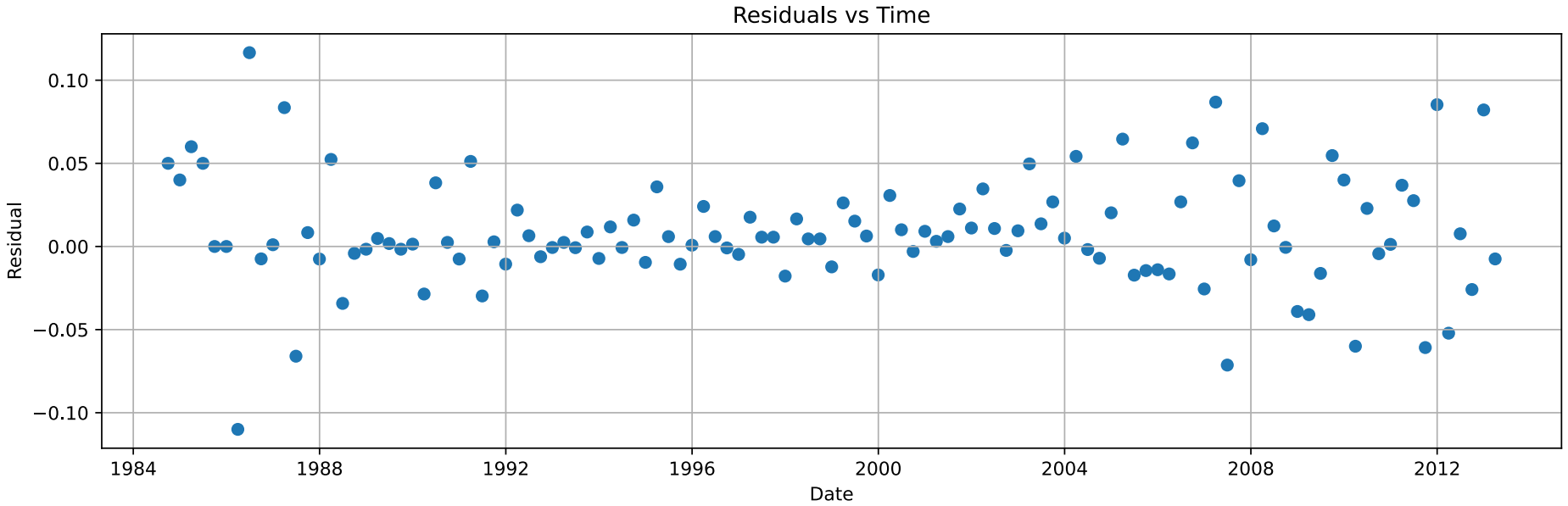
$B^4X_t = X_{t-4}$

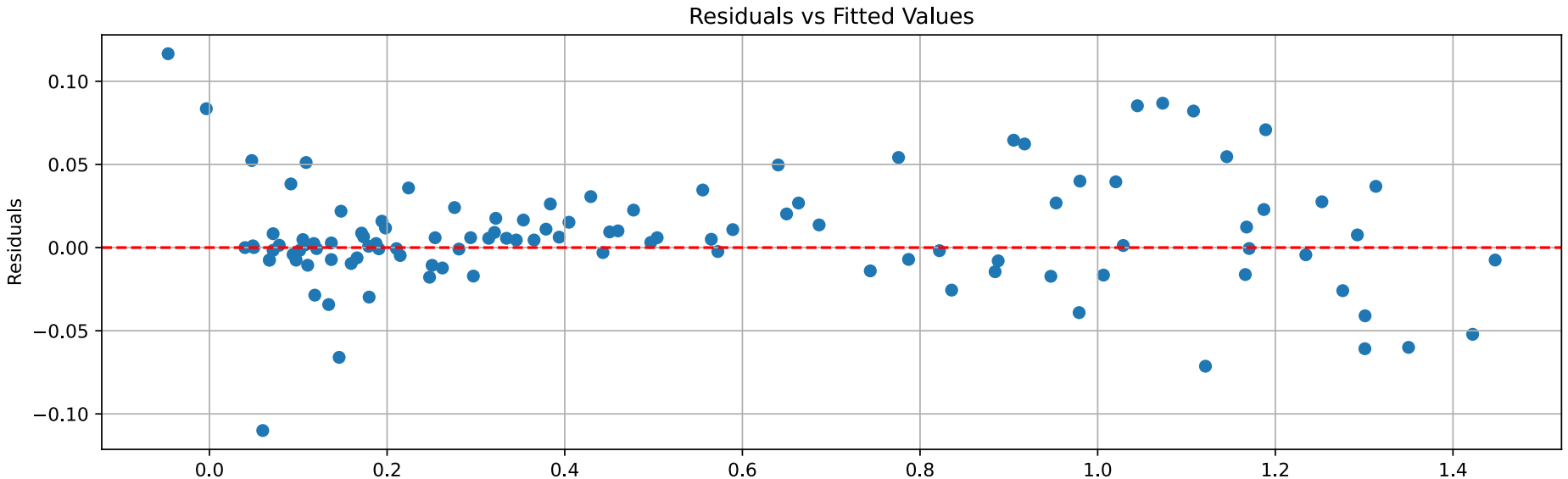
The Earnings per Share (EPS) of Johnson & Johnson



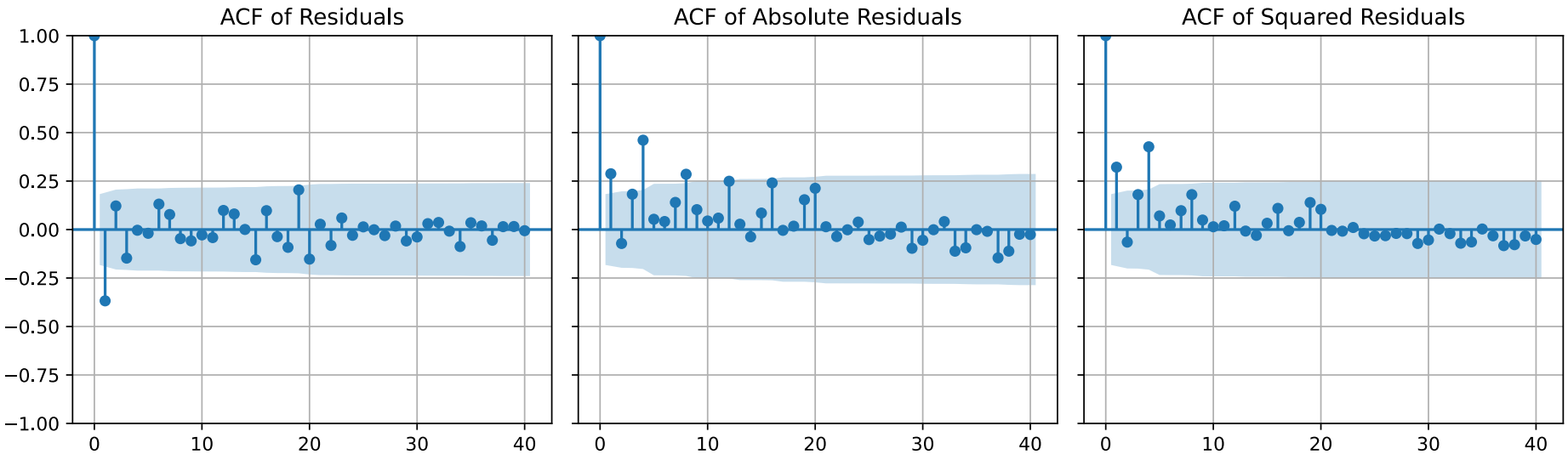
We fit a seasonal AR(1) model to the EPS series.

The Residuals of the Seasonal AR(1) Model





The ACF of the Residuals of the Seasonal AR(1) Model



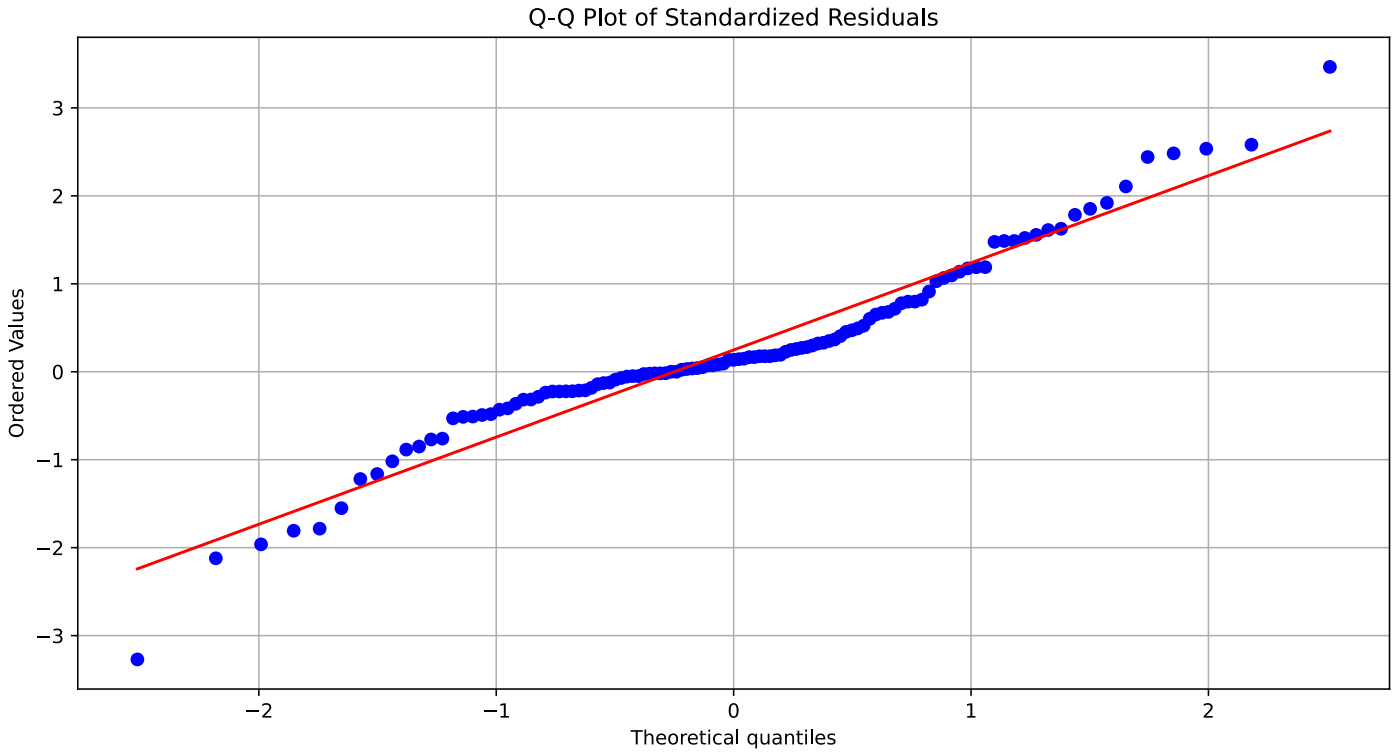
The Ljung-Box Test of the Residuals of the Seasonal AR(1) Model

Ljung-Box Test Results			
	Lag 10	Lag 15	Lag 20
Residuals			
Statistic	24.0607	29.6714	41.4760
p-value	0.0022	0.0053	0.0013
Absolute residuals			
Statistic	54.9417	64.7628	82.3944
p-value	0.0000	0.0000	0.0000
Squared residuals			
Statistic	44.9642	47.1785	53.2538
p-value	0.0000	0.0000	0.0000

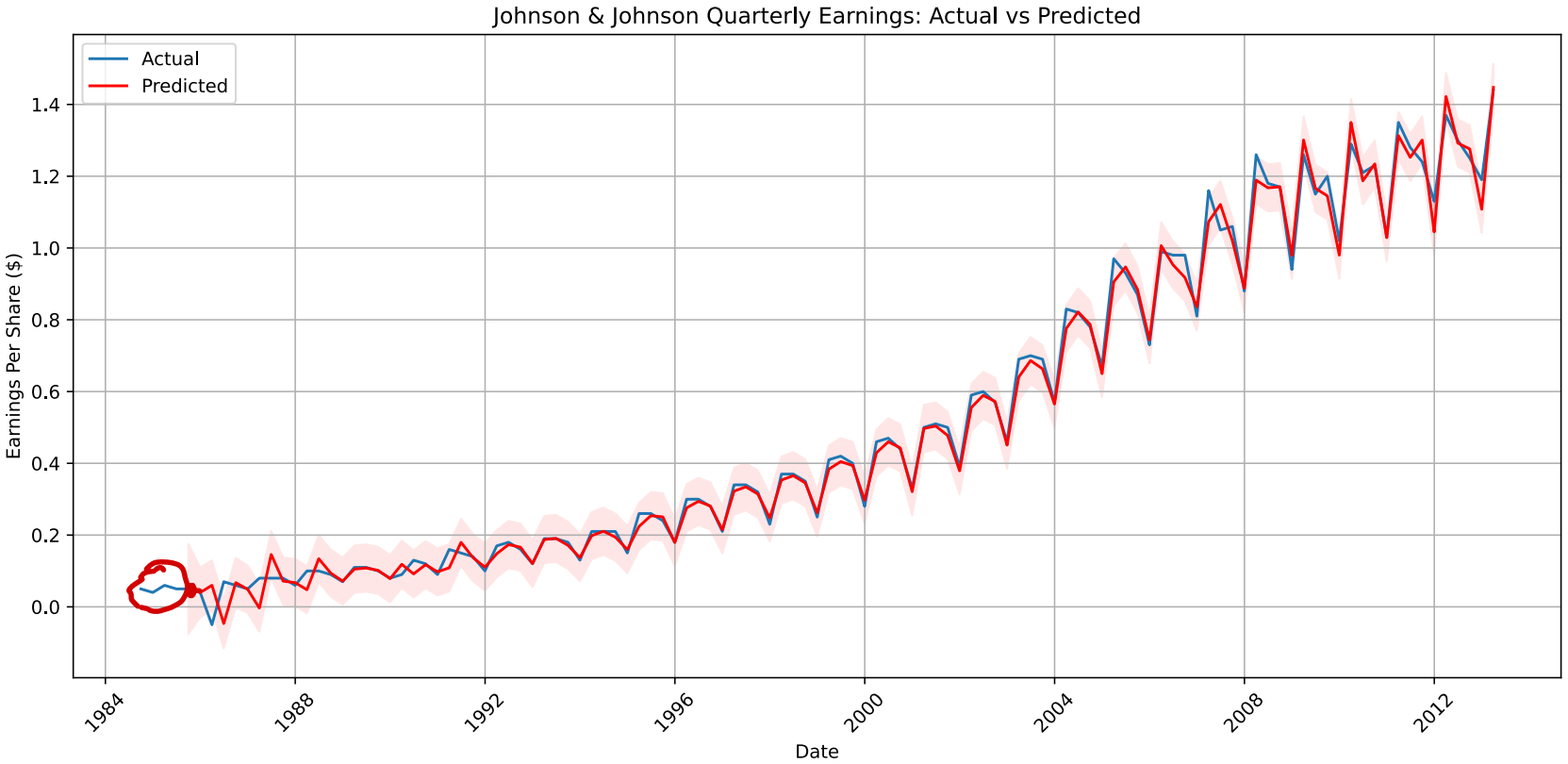
The QQ Plot of the Residuals of the Seasonal AR(1) Model

Jarque-Bera Test Results:

- Statistic: 13.9346
- p-value: 0.0009



Prediction of the Seasonal AR(1) Model



Outline

1. Stylized Facts of Financial Returns
2. Efficient Market Hypothesis and Statistical Models for Financial Returns
3. Time Series Models for Financial Returns
4. Volatility Modeling via GARCH Models
5. Applications and Extensions of GARCH Models

$r_t = \mu_t + \varepsilon_t$ white noise

μ_t: $E_{t-1}[r_t]$

$\varepsilon_t = \sigma_t \cdot \eta_t$

$\sigma_t = \sqrt{Var_{t-1}[r_t]}$

Volatility Estimation

Volatility means the degree of fluctuation of asset prices over time. It is important in many aspsects:

- Option pricing: volatility is a key parameter in the Black-Scholes option pricing model.
- Volatility is an important risk measure for risk management and evaluating risk-adjusted returns.

However, volatility is not directly observable. We need to estimate it from the observed price series.

- Implied volatility: the volatility implied by the option price, e.g., VIX.
- Realized volatility: estimated from the observed price series, computed as the sum of squared intraday returns for a particular day.
- Conditional volatility: estimated from dynamic models such as the ARCH and GARCH type models.

ARCH Models

The ARCH model has been introduced by Engle (1982)[2]:

$$\text{ARCH} = \text{AutoRegressive Conditional Heteroskedasticity}.$$

Example: an ARCH(1) process.

$$\begin{cases} X_t = Z_t \sigma_t, \\ \sigma_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2, \end{cases}$$

where Z_t is a sequence of i.i.d. random variables with mean 0 and variance 1, $\alpha_0 \geq 0$ and $0 \leq \alpha_1 < 1$.

- Heteroscedasticity refers to a time-varying conditional variance.
- The ARCH model assumes that the conditional variance is a linear function of the past squared returns.

[2]Robert F. Engle Nobel Prize 2003; Engle, R.F. (1982), AutoRegressive Conditional Heteroskedasticity with Estimates of the Variance of U.K. Inflation, Econometrica, 50, 987-1008.

Conditional Variance

The conditional variance:

$$\sigma_t^2 = \mathrm{Var}_{t-1}[X_t].$$

σ_t^2 is known based on the information up to time $t-1$.

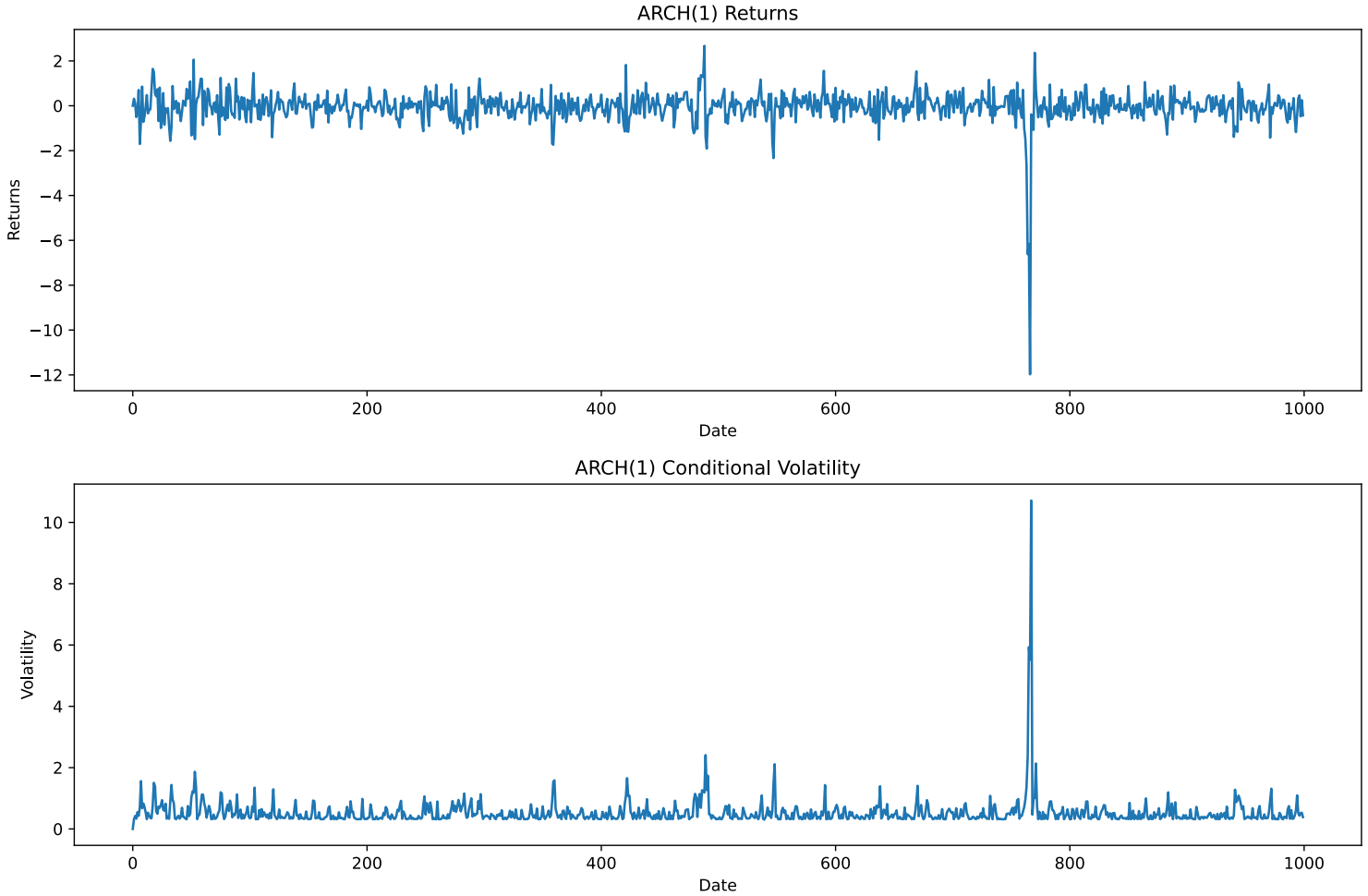
An equivalent formulation:

$$\begin{cases} X_t = Z_t\sqrt{h_t}, \\ h_t = \alpha_0 + \alpha_1 X_{t-1}^2, \end{cases}$$

where h_t is the conditional variance.

The Trajectory of an ARCH(1) Process

Set $\alpha_0 = 0.1$ and $\alpha_1 = 0.8$ and $Z_t \sim \mathcal{N}(0, 1)$.



The Properties of the ARCH Model: AR Representation

We have the following AR representation:

$$X_t^2 = \alpha_0 + \alpha_1 X_{t-1}^2 + v_t,$$

where $v_t = X_t^2 - \sigma_t^2$ satisfying that

$$\mathbb{E}_{t-1}[v_t] = 0.$$

The AR representation shows that the conditional variance of X_t^2 is an AR(1) process. Then

$$\mathbb{E}[X_t^2] = \frac{\alpha_0}{1-\alpha_1}, \ \mathrm{Cov}[X_t^2, X_{t-k}^2] = \alpha_1^k \mathrm{Var}[X_t^2] \neq 0,$$

which is known as the ARCH effect (corresponding to volatility clustering).

The ARCH(1) process is stationary if $\alpha_0 > 0$ and $0 \leq \alpha_1 < 1$.

The Properties of the ARCH Model: ACF of X_t

X_t is a martingale difference sequence with respect to the information up to $t-1$, i.e.,

$$\mathbb{E}_{t-1}[X_t] = 0.$$

The point prediction of X_t is the same as the ARMA prediction.

The ACF of X_t is given by:

$$\rho_k(X_t) = 0, \ \ k \geq 1.$$

This is consistent with the empirical fact that that the returns are not serially correlated.

The Properties of the ARCH Model: Kurtosis of X_t

The kurtosis of X_t is given by:

$$\text{Kurt}(X_t) = \frac{\mathbb{E}[X_t^4]}{\mathbb{E}[X_t^2]^2} = \frac{\mathbb{E}[Z_t^4]\mathbb{E}[\sigma_t^4]}{\mathbb{E}[Z_t^2]^2\mathbb{E}[\sigma_t^2]^2} \geq \text{Kurt}(Z_t).$$

The Properties of the ARCH Model: Fourth Moments

Assume $Z_t \sim \mathcal{N}(0,1)$.

The conditional fourth moment/kurtosis of X_t is given by:

$$\mathbb{E}[X_t^4|\mathcal{F}_{t-1}] = 3\sigma_t^4 \Rightarrow \text{Kurt}(X_t|\mathcal{F}_{t-1}) = 3.$$

The unconditional fourth moment of X_t is given by ($\alpha_1^2 < 1/3$):

$$\mathbb{E}[X_t^4] = \frac{3\alpha_0^2(1+\alpha_1)}{(1-3\alpha_1^2)(1-\alpha_1)} \Rightarrow \text{Kurt}(X_t) = \frac{3(1-\alpha_1^2)}{(1-3\alpha_1^2)} > 3.$$

The unconditional distribution of X_t is leptokurtic even if its conditional distribution is normal.

ARCH(p) Process

It is straightforward to extend the ARCH(1) model to the ARCH(p) model:

$$\begin{cases} X_t = Z_t \sigma_t, \\ \sigma_t^2 = \alpha_0 + \sum_{i=1}^{p} \alpha_i X_{t-i}^2, \end{cases}$$

where $\alpha_0 > 0$, $\alpha_i \geq 0$ for $i = 1, 2, \cdots, p$ and $\sum_{i=1}^{p} \alpha_i < 1$.

The ARCH(p) process is stationary if $\alpha_0 > 0$ and $\sum_{i=1}^{p} \alpha_i < 1$.

ARMA-ARCH Models

The return series can be modeled as an ARMA process with ARCH errors.

The returns can be decomposed as:

$$r_t = \mu_t + \varepsilon_t,$$

where ε_t follows an ARCH(p) process:

$$\varepsilon_t = \sigma_t Z_t,$$

where Z_t is a sequence of i.i.d. random variables with mean 0 and variance 1.

$$\mu_t = \mathbb{E}_{t-1}[r_t] = \mu_t(r_{:t-1}; \theta),$$
$$\sigma_t^2 = \mathrm{Var}_{t-1}[r_t] = \sigma_t^2(r_{:t-1}; \theta).$$

ARMA-ARCH Models (continued)

With an ARMA-ARCH specification, we can specify the conditional mean and variance of the returns.

$$\mu_t = \phi_0 + \sum_{i=1}^{p_1} \phi_i r_{t-i} + \sum_{j=1}^{q_1} \theta_j \varepsilon_{t-j},$$

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{p_2} \alpha_i \varepsilon_{t-i}^2.$$

Estimation of ARMA-ARCH Models

In most of the software, the parameters of the ARMA-ARCH models are estimated by maximizing the log-likelihood function assuming that $Z_t \sim \mathcal{N}(0,1)$.

After the estimation, we can solve for the conditional variance σ_t^2 and the residuals ε_t from the model.

The residuals ε_t can be used to test the model specification by the standardized residuals $\hat{Z}_t = \widehat{\varepsilon}_t / \widehat{\sigma}_t$.

- The Ljung-Box Q-statistics of $\hat{Z}_t$ can be used to check the adequacy of the mean equation.
- The Ljung-Box Q-statistics of $\hat{Z}_t^2$ can be used to check the adequacy of the volatility equation.
- The skewness, kurtosis, and QQ-plot of $\hat{Z}_t$ can be used to check the validity of the distribution assumption on Z_t.

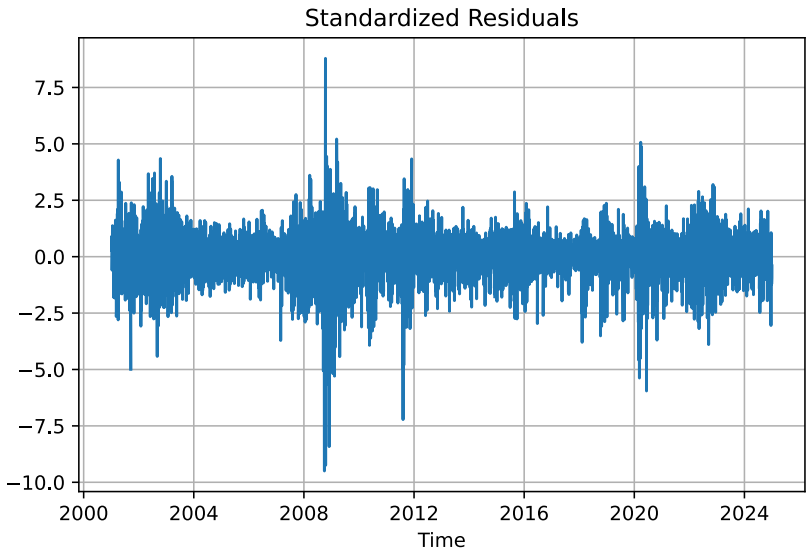
Fitting the AR(0)-ARCH(1) Model to SPX Returns

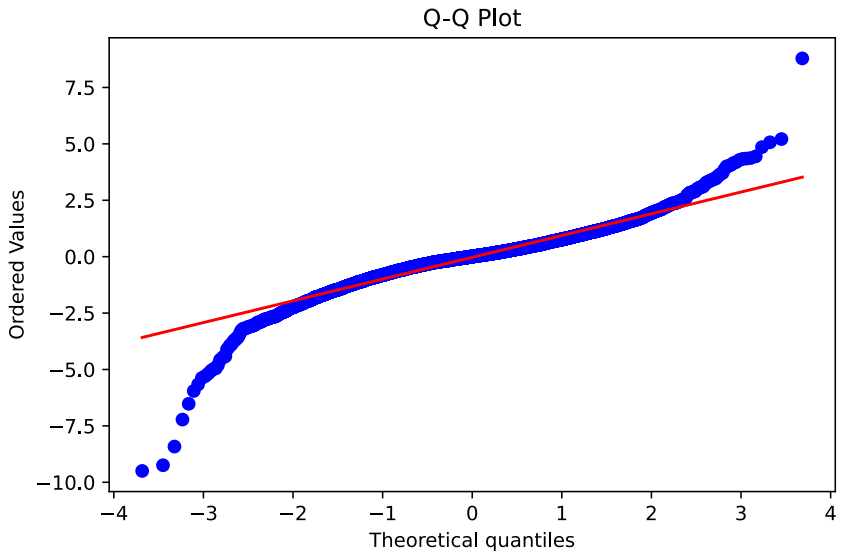
We fit the AR(0)-ARCH(1) model with constant mean to the SPX returns. In the estimation, the log returns have been scaled by 100 to ensure numerical stability.

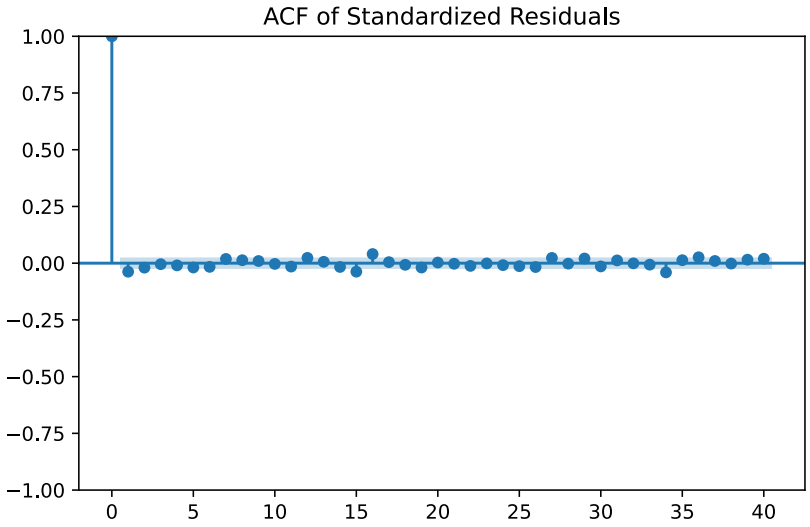
- Log-Likelihood: -9237.52.
- AIC: 18481.0.
- BIC: 18501.2.

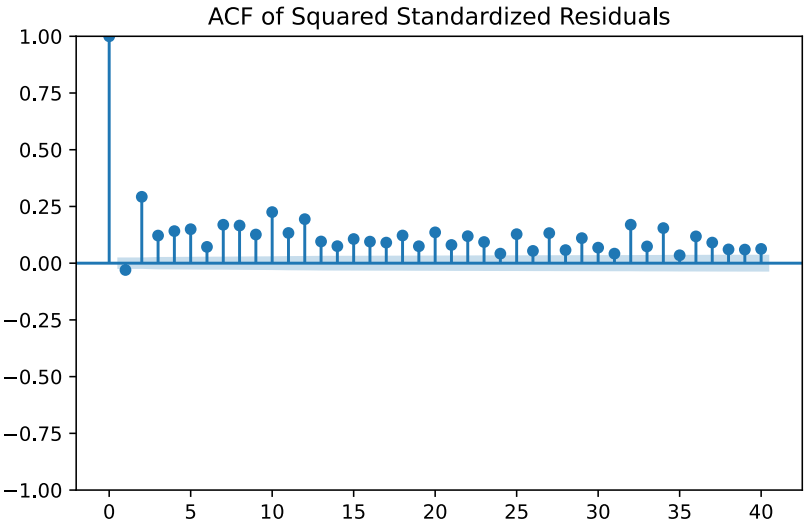
	Mean Model			
	coef	std err	t	$P > \|t\|$
Const	0.0562	1.629×10^{-2}	3.453	5.554×10^{-4}
	Volatility Model			
omega	0.9220	5.041×10^{-2}	18.289	1.010×10^{-74}
alpha[1]	0.4017	5.532×10^{-2}	7.262	3.803×10^{-13}

The AR(0)-ARCH(1) Model Diagnostics









The AR(0)-ARCH(1) Model Diagnostics (continued)

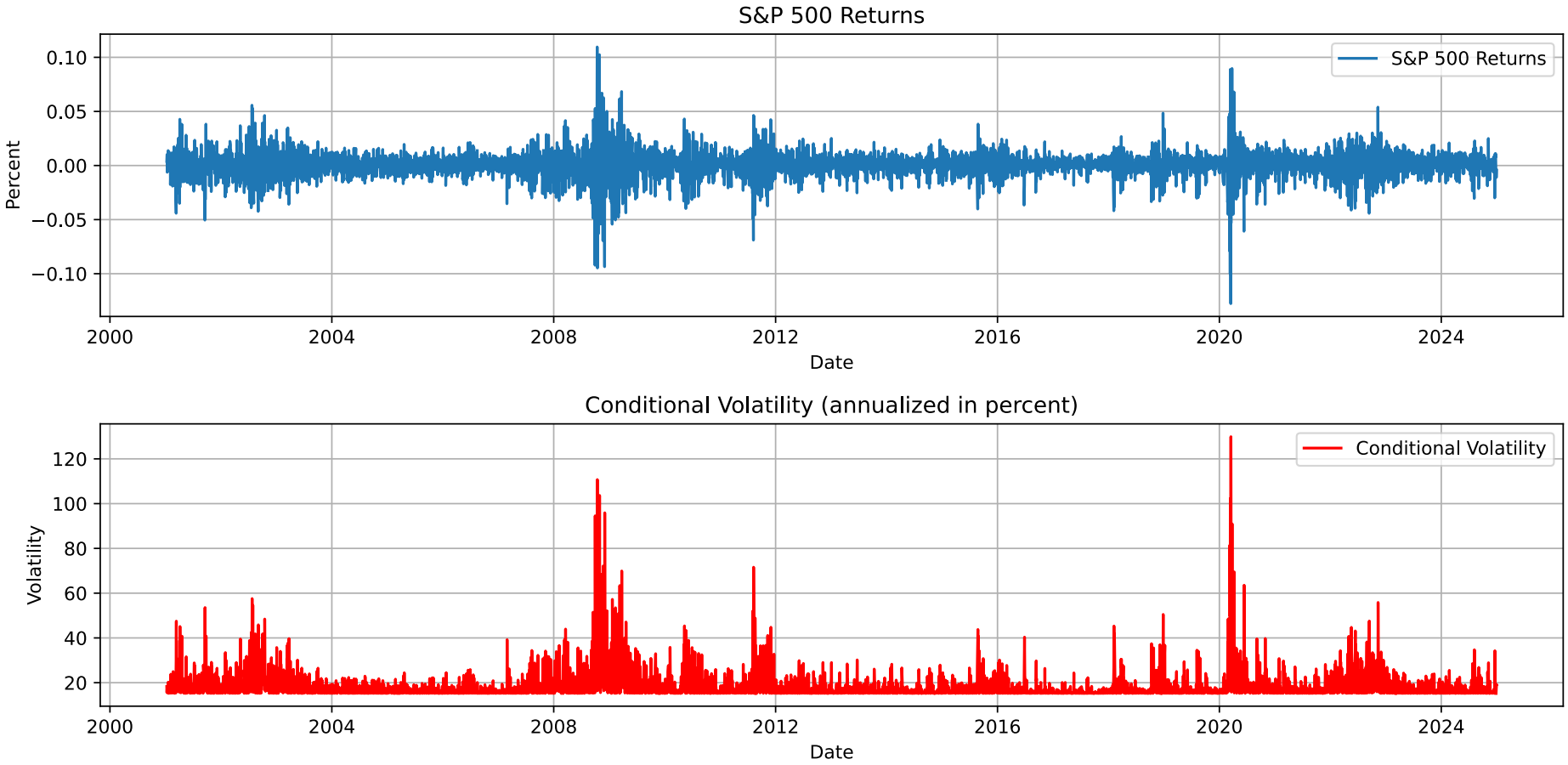
Ljung-Box Test Results			
	Lag 10	Lag 15	Lag 20
Residuals			
Statistic	19.09	34.08	46.59
p-value	0.0079	0.0007	0.0001
Absolute Residuals			
Statistic	1963.42	2674.30	3327.08
p-value	0.0000	0.0000	0.0000
Squared Residuals			
Statistic	1640.73	2133.21	2472.98
p-value	0.0000	0.0000	0.0000

The AR(0)-ARCH(1) Model Diagnostics (continued)

Jarque-Bera Test for Normality	
Statistic	15248.44
p-value	0.0000

Descriptive Statistics of Standardized Residuals	
Mean	-0.0287
Std Dev	0.9996
Skewness	-0.6004
Kurtosis	7.6960

Conditional Volatility of SPX Returns Estimated by the AR(0)-ARCH(1) Model



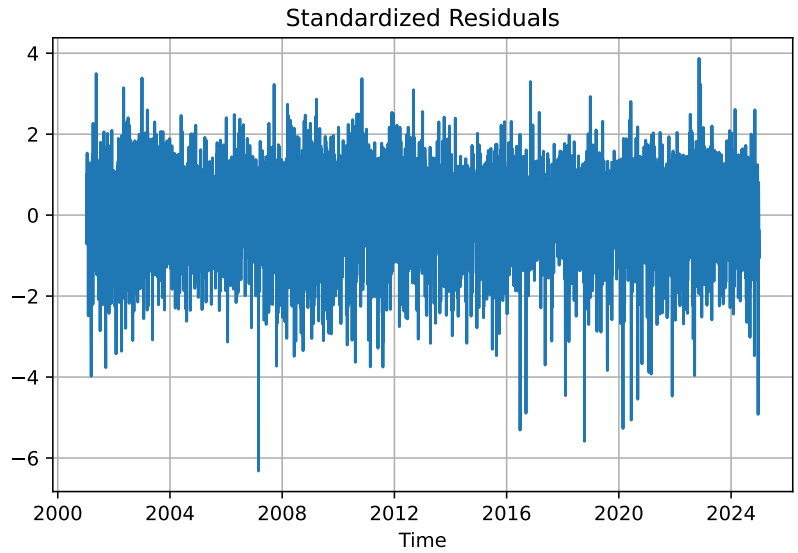
Fitting the AR(0)-ARCH(10) Model to SPX Returns

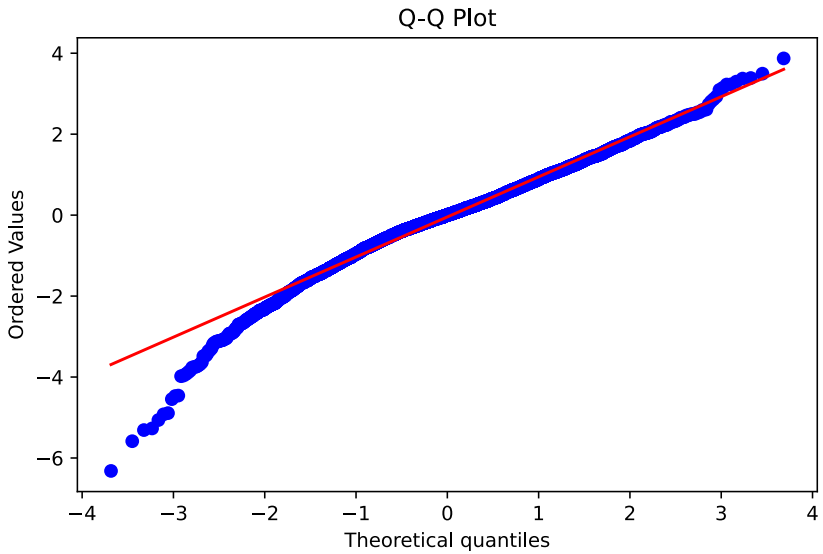
We fit the AR(0)-ARCH(10) model with constant mean to the SPX returns. In the estimation, the log returns have been scaled by 100 to ensure numerical stability.

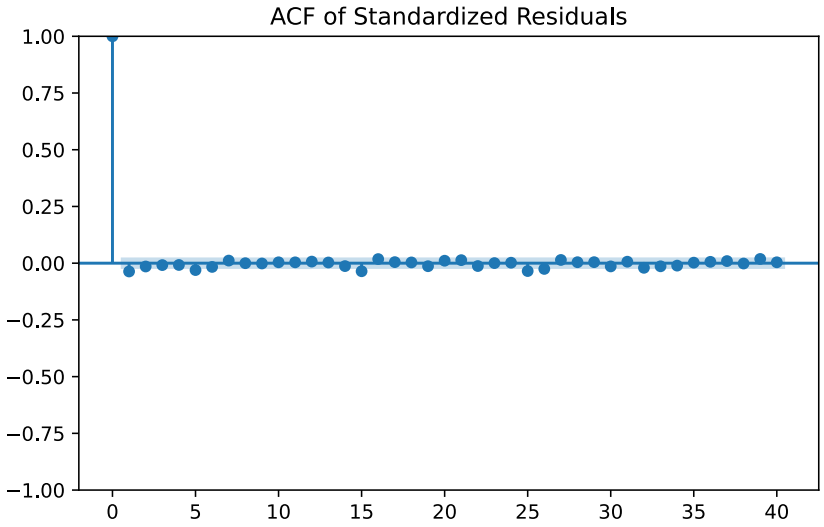
- Log-Likelihood: -8207.74.
- AIC: 16439.5.
- BIC: 16519.9.

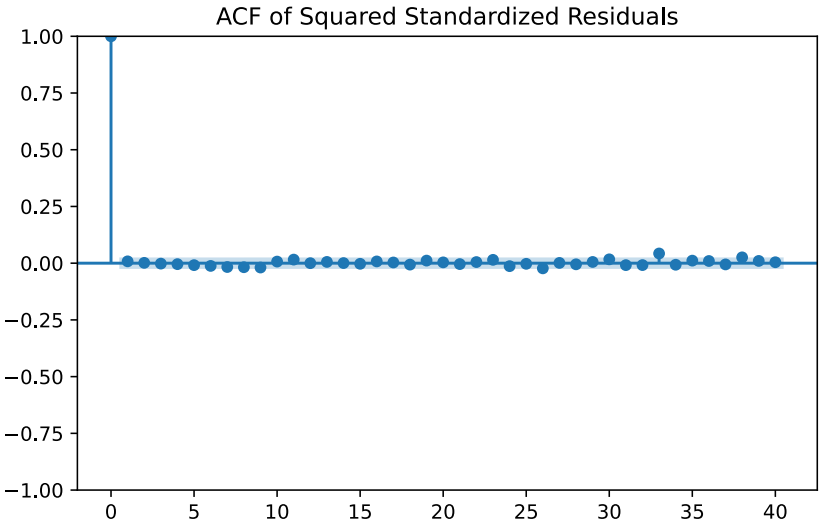
	Mean Model			
	coef	std err	t	$P > \|t\|$
Const	0.0630	1.022×10^{-2}	6.161	7.249×10^{-10}
	Volatility Model			
omega	0.2090	2.251×10^{-2}	9.283	1.650×10^{-20}
alpha[1]	0.0749	2.134×10^{-2}	3.510	4.489×10^{-4}
alpha[2]	0.1381	2.026×10^{-2}	6.817	9.305×10^{-12}
alpha[3]	0.1145	1.936×10^{-2}	5.912	3.385×10^{-9}
alpha[4]	0.1254	2.561×10^{-2}	4.896	9.770×10^{-7}
alpha[5]	0.0786	1.692×10^{-2}	4.646	3.382×10^{-6}
...	...	...	...	...
alpha[10]	0.0493	1.651×10^{-2}	2.985	2.837×10^{-3}

The AR(0)-ARCH(10) Model Diagnostics









The AR(0)-ARCH(10) Model Diagnostics (continued)

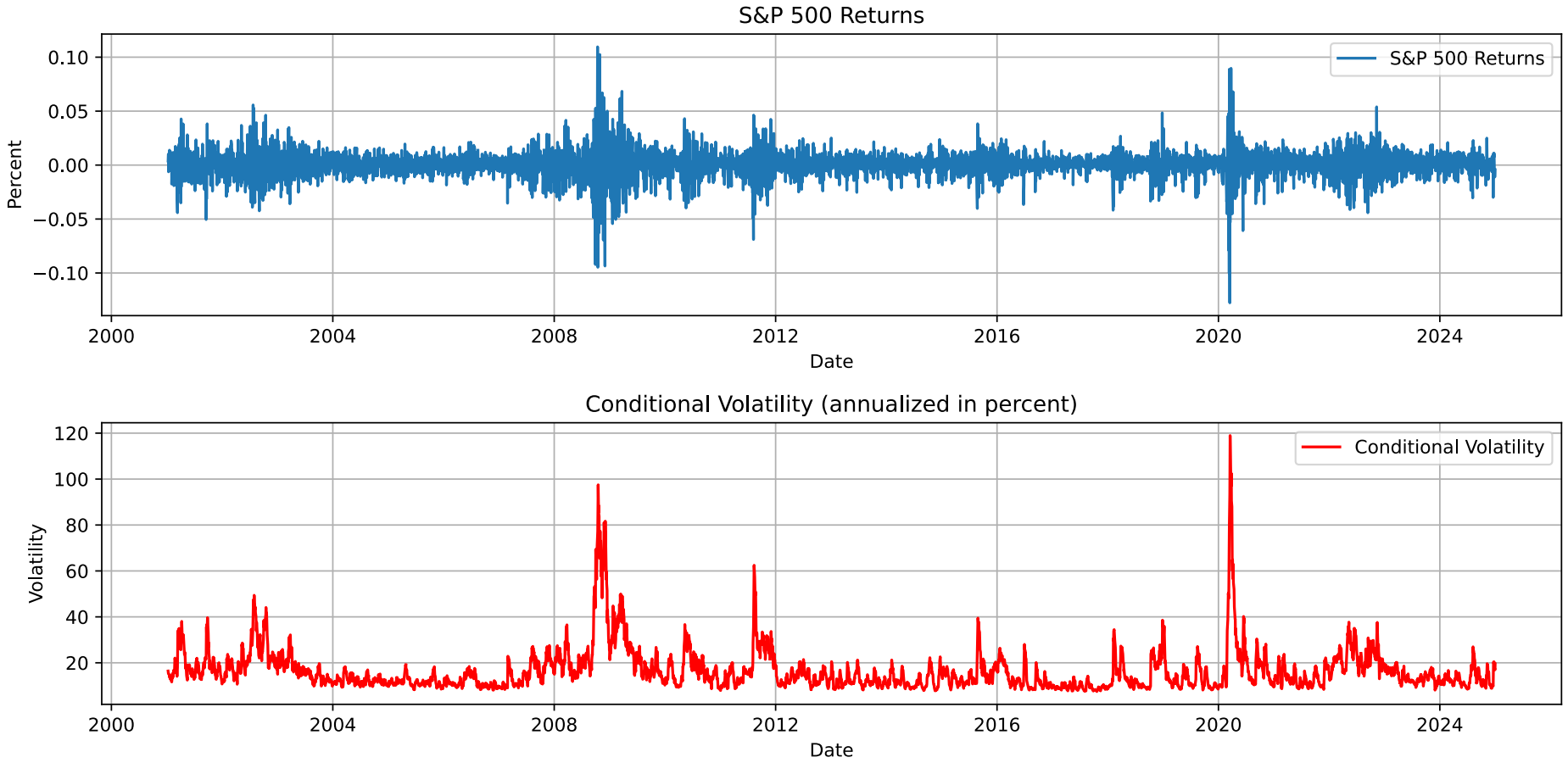
Ljung-Box Test Results			
	Lag 15	Lag 20	Lag 30
Residuals			
Statistic	27.36	31.32	47.24
p-value	0.0000	0.0001	0.0002
Absolute Residuals			
Statistic	16.17	25.85	43.89
p-value	0.0010	0.0011	0.0006
Squared Residuals			
Statistic	9.78	11.29	19.16
p-value	0.0206	0.1858	0.3822

The AR(0)-ARCH(10) Model Diagnostics (continued)

Jarque-Bera Test for Normality	
Statistic	948.15
p-value	0.0000

Descriptive Statistics of Standardized Residuals	
Mean	−0.0447
Std Dev	0.9990
Skewness	−0.5158
Kurtosis	1.6457

Conditional Volatility of SPX Returns Estimated by the AR(0)-ARCH(10) Model



GARCH Models

Due to the large persistence in volatility, ARCH models often require a large p to fit the data. A more parsimonious specification is provided by GARCH models[3].

GARCH = Generalized AutoRegressive Conditional Heteroskedasticity.

[3] Bollerslev, T. (1986), Generalized Autoregressive Conditional Heteroskedasticity. Journal of Econometrics, 31, 307-327.

GARCH Models (continued)

The asset return r_t can be decomposed as:

$$\begin{cases} r_t = \mu_t + \varepsilon_t, \\ \varepsilon_t = \sigma_t Z_t, \end{cases}$$

where Z_t is a sequence of i.i.d. random variables with mean zero and variance 1, and,

$$\mu_t = \mathbb{E}[r_t|\mathcal{F}_{t-1}],$$
$$\sigma_t^2 = \mathrm{Var}[r_t|\mathcal{F}_{t-1}].$$

The GARCH(p, q) process is given by:

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2,$$

where $\omega > 0$, $\alpha_i \geq 0$, $\beta_j \geq 0$, and $\sum_{i=1}^{p} \alpha_i + \sum_{j=1}^{q} \beta_j < 1$.

GARCH Models (continued)

The parameters α_i are called ARCH parameters and β_j GARCH parameters.

From a practical point, GARCH$(1,1)$ specifications are generally sufficient to capture the dynamics of the conditional variance and higher-order lags are not required.

The GARCH$(1,1)$ process is given by:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2.$$

The overall impact of a shock can be decomposed into a contemporaneous effect, which depends on α, and a persistence effect, which depends on β.

GARCH Models (continued)

One often observes that:

- The sum of the estimates of α and β are generally close (but below 1).
- The estimate of β is generally greater than the one of α.
- The estimate of β is generally larger than 0.90 for daily returns and the estimate of α is below 0.1.

Properties of GARCH Models

The main properties of a GARCH process are similar to those of an ARCH process.

- The process ε_t^2 has an ARMA representation.
- The process ε_t is a martingale difference.
- The process ε_t is a stationary process under some conditions on the parameters α and β.
- The process ε_t is (unconditionally) homoscedastic.
- The process ε_t is conditionally heteroscedastic.
- The (marginal) distributions of r_t and Z_t are leptokurtic.
- If Z_t has a normal distribution, the conditional distributions of r_t is normal.

ARMA Representation of GARCH Models

It is not difficult to show that:

$$\varepsilon_t^2 = \omega + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i)\varepsilon_{t-i}^2 + v_t - \sum_{j=1}^{q} \beta_j v_{t-j},$$

where $v_t = \varepsilon_t^2 - \sigma_t^2$ satisfies that $\mathbb{E}_{t-1}[v_t] = 0$.

Unconditional Moments of GARCH Models

The unconditional mean and variance of ε_t is given by:

$$\mathbb{E}[\varepsilon_t] = 0,$$
$$\text{Var}[\varepsilon_t] = \frac{\omega}{1 - \sum_{i=1}^{p} \alpha_i - \sum_{j=1}^{q} \beta_j}.$$

Therefore, the variance equation can be written as:

$$\sigma_t^2 = \text{Var}[\varepsilon_t] + \sum_{i=1}^{p} \alpha_i (\varepsilon_{t-i}^2 - \text{Var}[\varepsilon_t]) + \sum_{j=1}^{q} \beta_j (\sigma_{t-j}^2 - \text{Var}[\varepsilon_t]).$$

Kurtosis of GARCH Models

Assuming that Z_t is a standard normal random variable, Bollerslev (1986) shows that the kurtosis coefficient of a GARCH(1,1) process is equal to:

$$\text{Kurt}(\varepsilon_t) = \frac{3(1-(\alpha+\beta)^2)}{1-(\alpha+\beta)^2-2\alpha^2},$$

when $(\alpha+\beta)^2+2\alpha^2<1$.

The kurtosis coefficent is infinite if $(\alpha+\beta)^2+2\alpha^2 \geq 1$ which is the case for many practical applications.

Estimation of GARCH Models

The set of parameters θ of an ARMA-GARCH model is estimated by Maximum Likelihood (ML) or Quasi Maximum Likelihood (QML).
When the model is estimated by ML, the most often used distributions for Z_t are:

1. The normal distribution, $Z_t \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$.
2. The Student t-distribution, $Z_t \overset{i.i.d.}{\sim} t(\nu)$, which is symmetric and leptokurtic (if ν is "small").
3. The skewed Student t-distribution, $Z_t \overset{i.i.d.}{\sim}$ Skewed-$t(\delta,\nu)$, which is asymmetric (if $\delta \neq 1$) and leptokurtic (if ν is "small").
4. The Generalized Error Distribution (GED), $Z_t \overset{i.i.d.}{\sim}$ GED(ν), which is symmetric and leptokurtic (if $\nu < 2$).

Estimation of GARCH Models (continued)

Suppose the density of Z_t is known as $f(\cdot)$. Then, the log-likelihood function of the GARCH model is given by:

$$\ell(\theta) = \sum_{t=\nu+1}^{T} \log\left(\frac{1}{\sigma_t} f\left(\frac{\varepsilon_t}{\sigma_t}\right)\right) = \sum_{t=\nu+1}^{T} \left(-\log \sigma_t + \log f\left(\frac{\varepsilon_t}{\sigma_t}\right)\right),$$

where $\frac{\varepsilon_t}{\sigma_t}$ is the standardized innovation.
When $Z_t \sim \mathcal{N}(0,1)$, the log-likelihood function is given by:

$$\begin{aligned}\ell(\theta) &= \sum_{t=\nu+1}^{T} \left(-\log \sigma_t + \log f\left(\frac{\varepsilon_t}{\sigma_t}\right)\right) \\ &= \sum_{t=\nu+1}^{T} \left(-\log \sigma_t - \frac{\varepsilon_t^2}{2\sigma_t^2} + \log \frac{1}{\sqrt{2\pi}}\right).\end{aligned}$$

Why to consider non-normal distributions for Z_t?

Why consider non-Gaussian distributions for the innovation Z_t?

1. The use of a leptokurtic distribution for Z_t allows to increase the kurtosis of r_t:

$$\text{kurtosis of a GARCH process} = \text{kurtosis from model dynamics} + \text{kurtosis of } Z_t$$

   To reproduce the high kurtosis observed in financial returns, the kurtosis generated by the model dynamics alone is not sufficient. This motivates the use of leptokurtic distributions for Z_t like Student-t or GED.

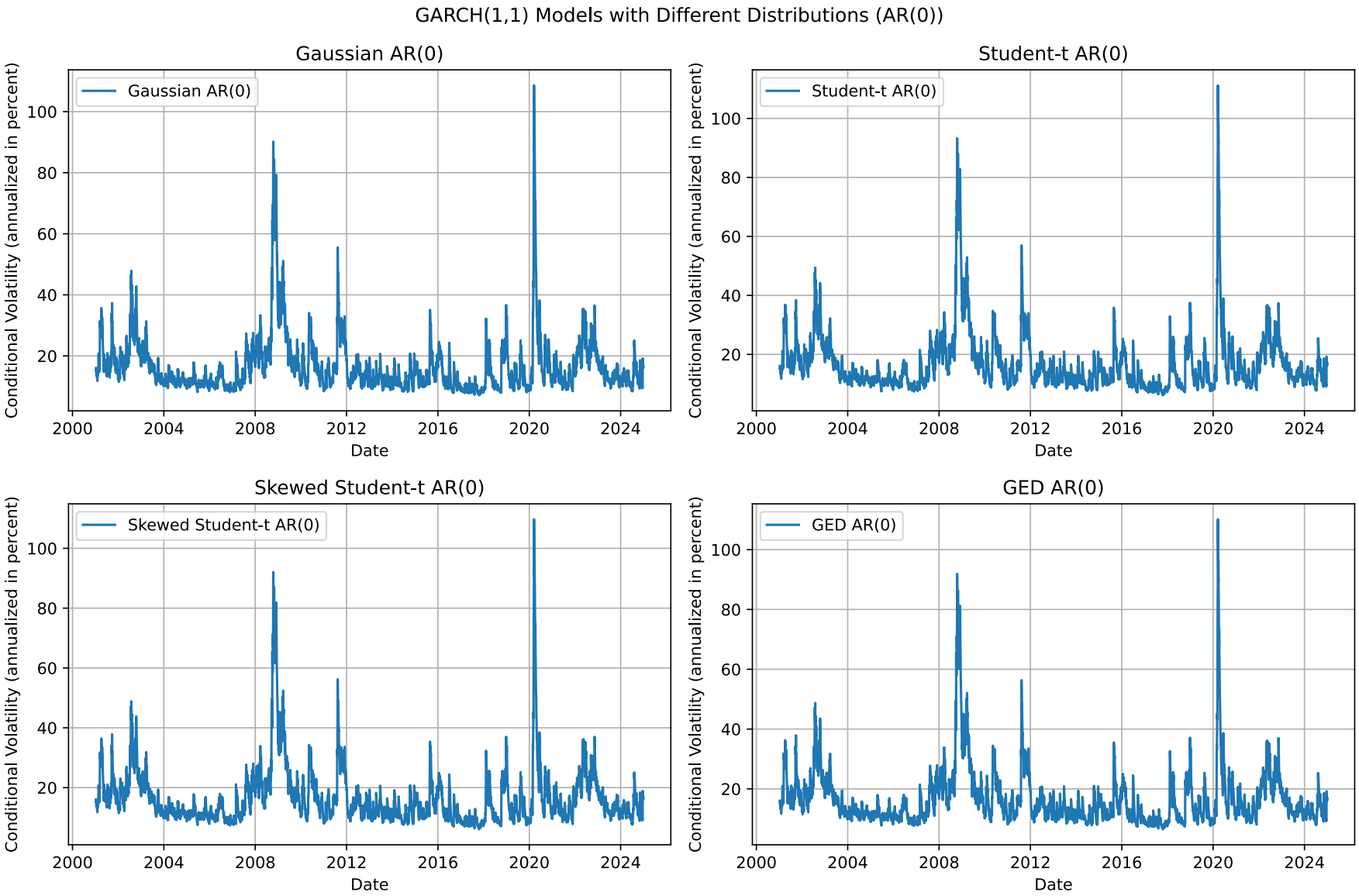
2. The use of a skewed distribution for Z_t allows to reproduce the skewness observed in financial returns:

$$\text{skewed distribution for } Z_t \Rightarrow \text{skewed distribution for } r_t$$

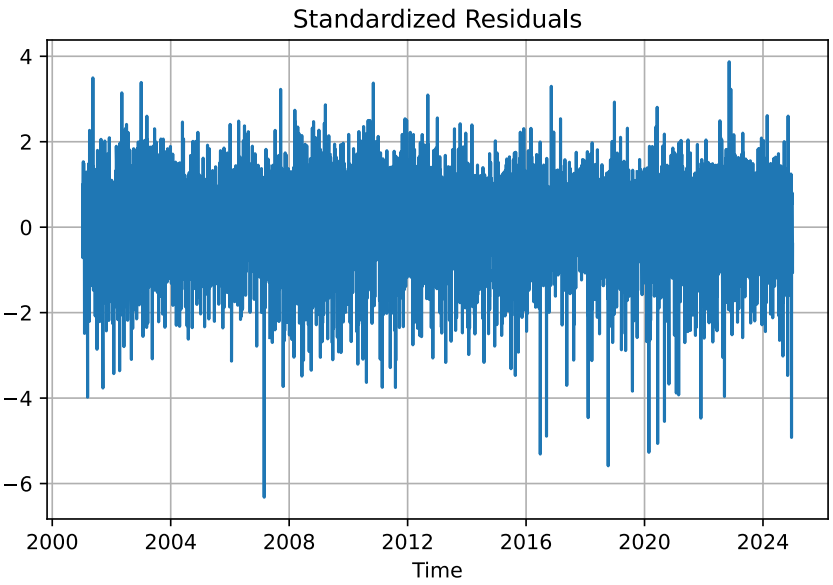
Fitting GARCH Models to SPX Returns

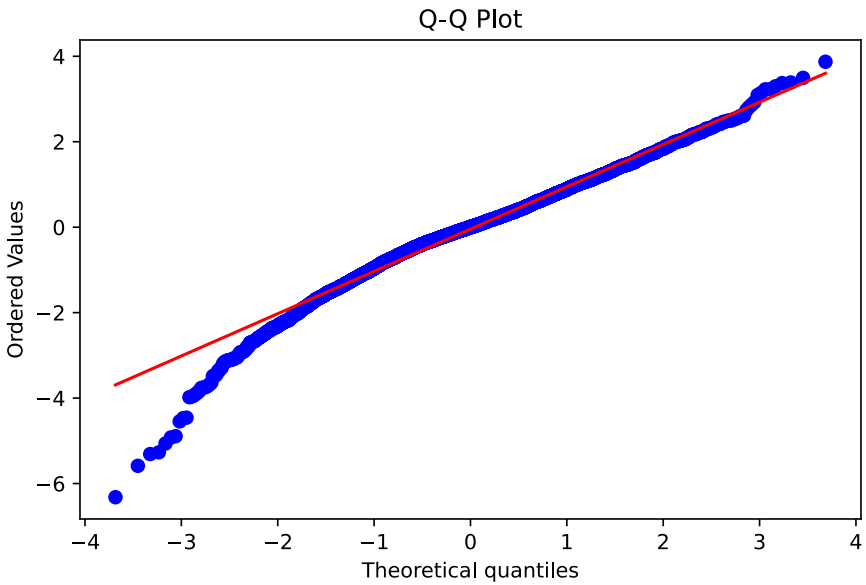
	Gaussian	Student-t	Skewed Student-t	GED
Log-Likelihood	−8202.17	−8068.88	−8047.69	−8061.94
AIC	16412.3	16147.8	16107.4	16133.9
BIC	16439.2	16181.3	16147.6	16167.4
Mean Model				
Const	0.0629	0.0796	0.0602	0.0751
t-stat	6.088	8.605	6.066	9.392
p-value	1.144×10^{-9}	7.636×10^{-18}	1.308×10^{-9}	5.869×10^{-21}
Volatility Model				
ω	0.0240	0.0148	0.0143	0.0183
t-stat	4.953	4.699	4.774	5.047
p-value	7.317×10^{-7}	2.620×10^{-6}	1.806×10^{-6}	4.479×10^{-7}
α_1	0.1197	0.1225	0.1191	0.1215
t-stat	9.794	10.992	11.307	10.819
p-value	1.193×10^{-22}	4.169×10^{-28}	1.216×10^{-29}	2.795×10^{-27}
β_1	0.8616	0.8731	0.8742	0.8677
t-stat	66.455	81.728	83.809	76.779
p-value	0.000	0.000	0.000	0.000

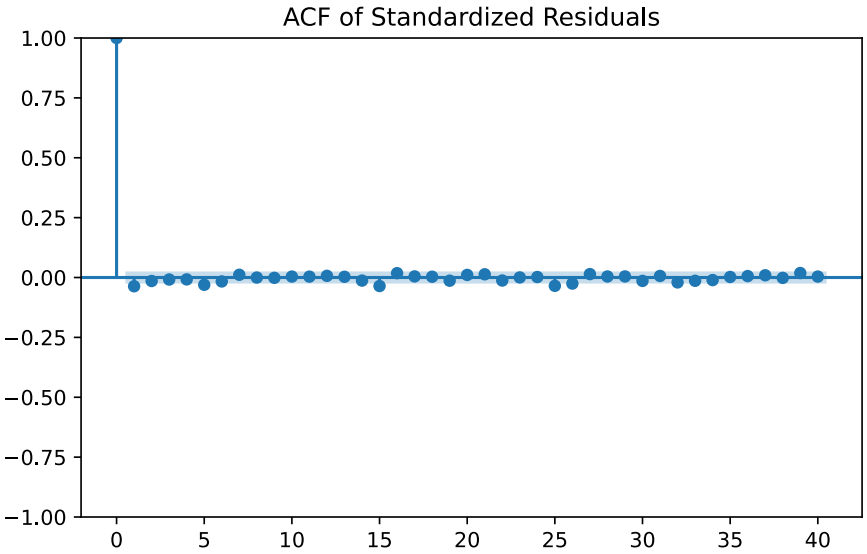
Fitting GARCH Models to SPX Returns (continued)

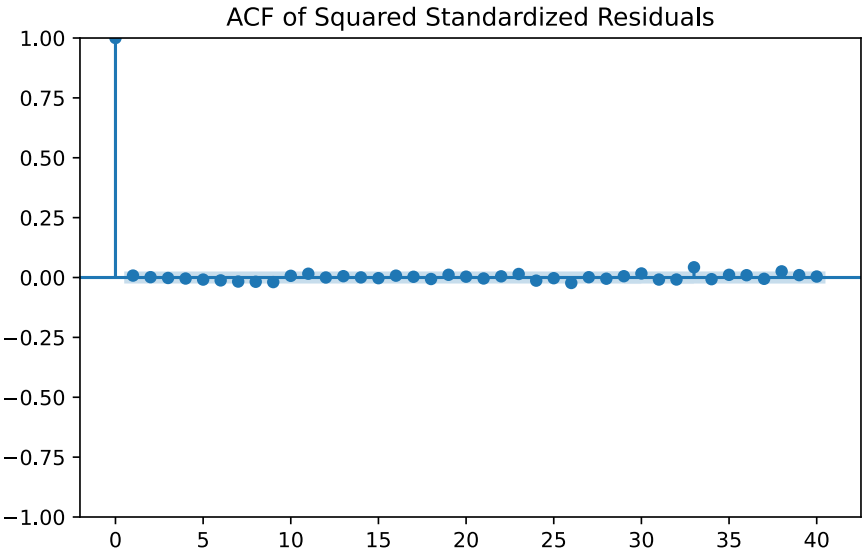


The Gaussian AR(0)-GARCH(1,1) Model Diagnostics









The Ljung-Box Test for the Gaussian AR(0)-GARCH(1,1) Model Residuals

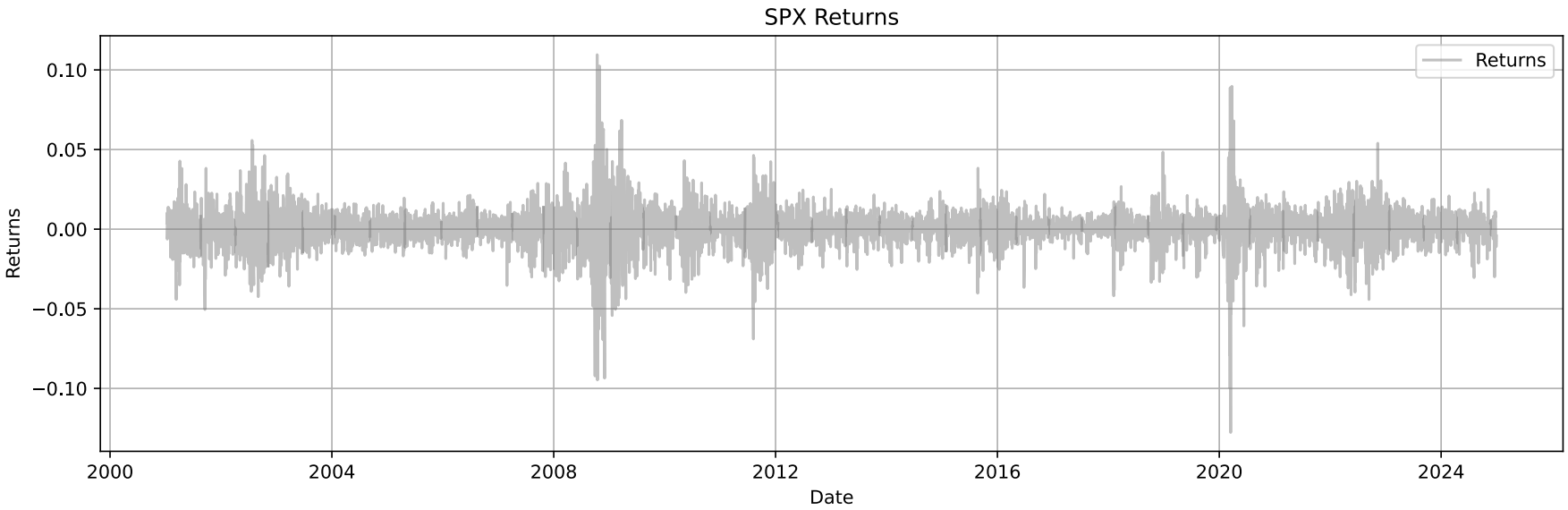
	Lag 10	Lag 20	Lag 30
Residuals			
Statistic	18.78	31.12	46.91
p-value	0.0046	0.0130	0.0072
Absolute Residuals			
Statistic	34.93	40.79	52.95
p-value	0.0000	0.0006	0.0014
Squared Residuals			
Statistic	14.96	18.74	29.46
p-value	0.0206	0.2824	0.2906

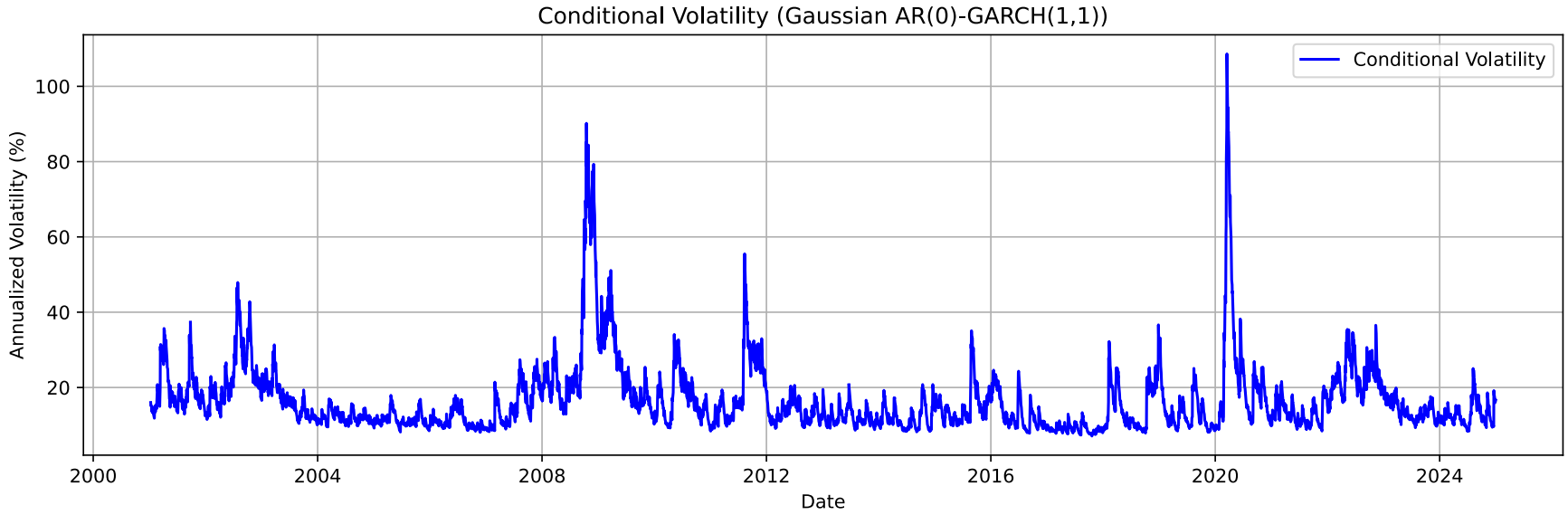
The Jarque-Bera Test for the Gaussian AR(0)-GARCH(1,1) Model Residuals

Jarque-Bera Test for Normality	
Statistic	999.30
P-value	0.0000

Descriptive Statistics of Standardized Residuals	
Mean	−0.0446
Std Dev	0.9989
Skewness	−0.5327
Kurtosis	1.6854

Fitting GARCH Models to SPX Returns (continued)





Outline

1 Stylized Facts of Financial Returns

2 Efficient Market Hypothesis and Statistical Models for Financial Returns

3 Time Series Models for Financial Returns

4 Volatility Modeling via GARCH Models

5 Applications and Extensions of GARCH Models

Volatility Forecasting

Volatility forecasting is an important application of GARCH models.

We note that:

$$\varepsilon_{t+h}^2 = \omega + \sum_{i=1}^{\max(p,q)} \alpha_i \varepsilon_{t+h-i}^2 + v_{t+h} + \sum_{j=1}^{q} \beta_j v_{t+h-j},$$

where $v_t = \varepsilon_t^2 - \sigma_t^2$. Then,

$$\sigma_{t+h}^2 = \mathbb{E}_t[\varepsilon_{t+h}^2] = \omega + \sum_{i=1}^{\max(p,q)} \alpha_i \mathbb{E}_t[\varepsilon_{t+h-i}^2] + \sum_{j=1}^{q} \beta_j \mathbb{E}_t[v_{t+h-j}],$$

where

$$\mathbb{E}_t[\varepsilon_{t+h-i}^2] = \begin{cases} \sigma_{t+h-i}^2 & \text{if } i < h \\ \varepsilon_{t+h-i}^2 & \text{if } i \geq h \end{cases}, \quad \mathbb{E}_t[v_{t+h-j}] = \begin{cases} 0 & \text{if } j < h \\ v_{t+h-j} & \text{if } j \geq h \end{cases}.$$

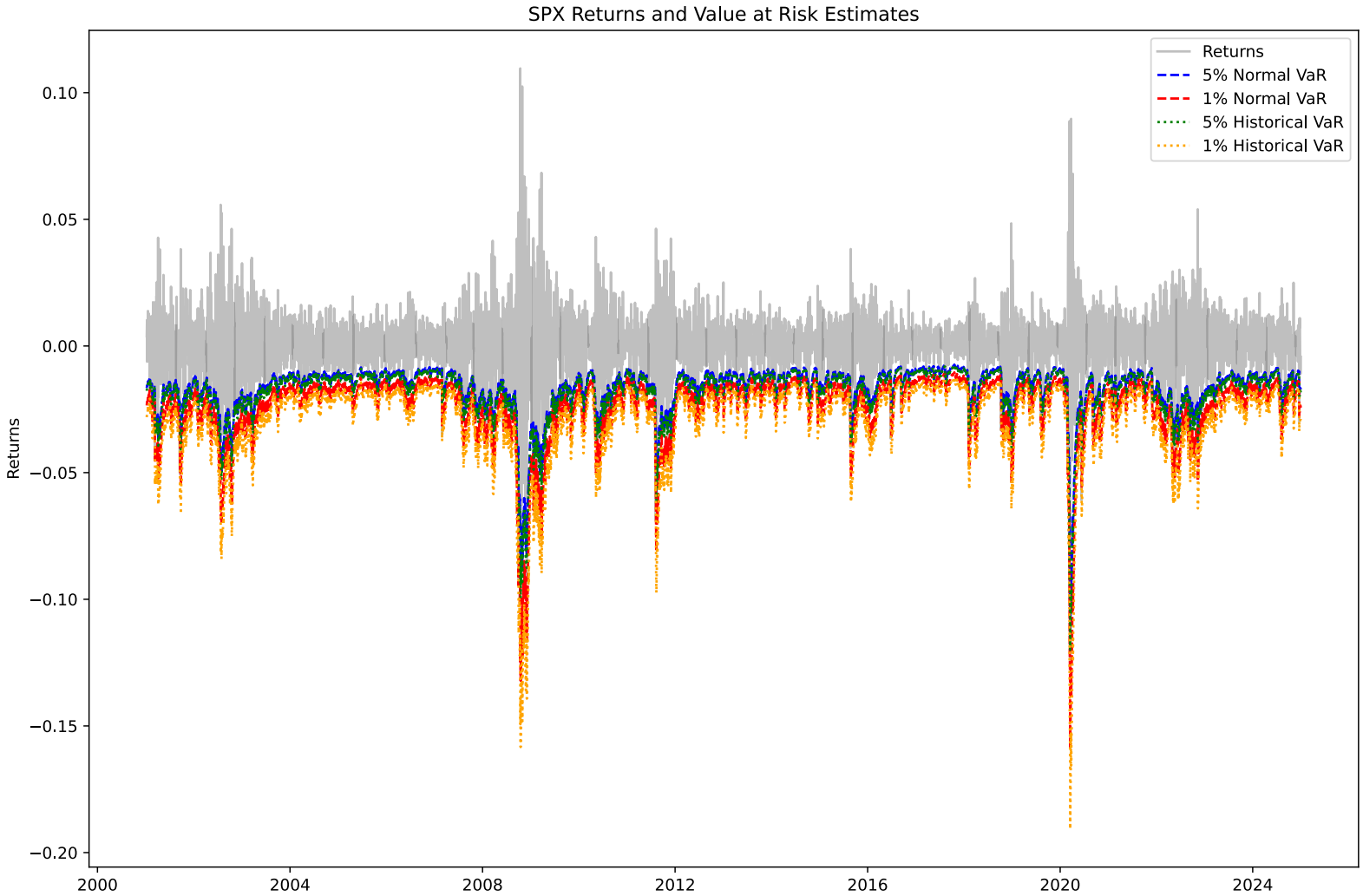
Forecasting Value-at-Risk (VaR) with GARCH Models

Suppose you are holding SPX, the VaR can be computed as:

$$\text{VaR}_\alpha = \sigma_{t+1} \times F_Z^{-1}(\alpha),$$

where $F_Z^{-1}(\alpha)$ can be estimated either by the inverse of the cumulative distribution function of the assumed distribution for Z_t or by empirical quantiles.

Forecasting Value-at-Risk (VaR) with GARCH Models (continued)



GARCH Extensions

Some relevant extensions of the GARCH model have been proposed in order to accommodate particular features of financial series (asymmetry, leverage effect, etc.).

Among others, GARCH models have been refined by introducing:

1. Asymmetric responses to negative and positive innovations to handle the observed asymmetry in the reaction of conditional volatility to the arrivals of news.
2. Persistence.
3. Long-memory (the dependency for a large number of lags).

GARCH Extensions (continued)

The following GARCH models are often encountered in the empirical financial literature as well as in the industry:

1. Asymmetric GARCH models: Exponential GARCH model (EGARCH), Threshold GARCH model (TGARCH), GJR model;
2. Integrated GARCH model (IGARCH).
3. Long-memory GARCH model (LMGARCH) or Fractionally integrated GARCH model (FIGARCH).

GARCH Extensions: IGARCH

Integrated GARCH (IGARCH)[4] models are a special case of GARCH models where the sum of the ARCH and GARCH parameters is equal to 1.

Example: IGARCH(1, 1):

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + (1-\alpha)\sigma_{t-1}^2,$$

The IGARCH model has several important properties:

1. The impact of past squared shocks $\varepsilon_{t-k}^2 - \sigma_{t-k}^2$ in the ARMA representation is persistent.
2. The unconditional variance of ε_t is not defined under an IGARCH(1, 1) model.
3. A special case of the IGARCH(1, 1) is the RiskMetrics volatility model defined as: $\sigma_t^2 = \lambda\varepsilon_{t-1}^2 + (1-\lambda)\sigma_{t-1}^2$, which is commonly used to compute Value-at-Risk.

[4] Nelson, D. B. (1990), Stationarity and persistence in the GARCH(1, 1) model, Econometric Theory, 6, 318-334.

GARCH Extensions: GARCH-M

The return of a security may depend on its volatility. To model such a phenomenon, one may consider the GARCH-M[5] model, where M stands for GARCH in mean.

$$r_t = c + \delta\sigma_t^2 + \varepsilon_t \qquad \text{(Conditional variance effect)}$$
$$r_t = c + \delta\sigma_t + \varepsilon_t \qquad \text{(Conditional volatility effect)}$$
$$r_t = c + \delta \ln \sigma_t + \varepsilon_t \qquad \text{(Log-linear specification)}$$

The parameter δ is called the risk premium parameter. A positive δ indicates that the return is positively related to its past volatility.

[5] Engle, R., Lilien, D., and R. Robins (1987). Estimating Time Varying Risk Premia in the Term Structure: The ARCH-M Model. Econometrica, 55(2), 391-407.

GARCH Extensions: Asymmetric GARCH Models

Asymmetric GARCH models:

- The GARCH model assumes that positive and negative shocks have the same effects on volatility because it depends on the square of the previous shocks.
- In practice, the return of a financial asset responds differently to positive and negative shocks.
- The GARCH model does not allow to capture the leverage effect.

As asset prices decline, companies become more leveraged (debt to equity ratios increase) and riskier, and hence their stock prices become more volatile.

On the other hand, when stock prices become more volatile, investors demand high returns and hence stock prices go down.

GARCH Extensions: Asymmetric GARCH Models (continued)

The asymmetric GARCH models are designed to capture the nonlinearities of the conditional variance dynamics, including the leverage effect.

Many asymmetric GARCH models have been proposed: GJR-GARCH, TGARCH, EGARCH, APARCH, VSGARCH, QGARCH, LSTGARCH, ANSTGARCH, etc.

One of the most often used asymmetric models is the GJR-GARCH[6] model.

[6] Glosten, L., Jagannathan, R., and D. Runkle, D. (1993). On the Relation between the Expected Value and the Volatility of the Nominal Excess Return on Stocks. The Journal of Finance, 48(5), 1779-1801.

GARCH Extensions: Asymmetric GARCH Models (continued)

The GJR-GARCH model is given by:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \gamma\varepsilon_{t-1}^2 \cdot I(\varepsilon_{t-1} < 0) + \beta\sigma_{t-1}^2.$$

The impact of past shocks on the conditional variance is:

$$\frac{\partial\sigma_t^2}{\partial\varepsilon_{t-1}^2} = \begin{cases} \alpha + \gamma & \text{if } \varepsilon_{t-1} < 0 \\ \alpha & \text{otherwise} \end{cases}$$

A leverage effect implies that $\gamma > 0$, i.e., the increase in volatility caused by a negative shock is larger than the appreciation due a positive shock of the same magnitude.

GARCH Extensions: TGARCH

The TGARCH[7] model is given by:

$$\sigma_t = \omega + \alpha_+ \varepsilon_{t-1} I(\varepsilon_{t-1} \geq 0) + \alpha_- \varepsilon_{t-1} I(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1},$$

where $I(\cdot)$ is the indicator function.

The TGARCH allows to capture an asymmetry between positive and negative shocks, as the leverage effect implies that $|\alpha_-| > |\alpha_+|$, i.e. the increase in volatility caused by a negative return is larger than the appreciation due a positive return of the same magnitude.

[7] Zakoian J.M. (1994), Threshold Heteroskedastic Models. Journal of Economic Dynamic and Control, 18, 931-955, 1994.

GARCH Extensions: EGARCH

The EGARCH, where "E" stands for Exponential, is an asymmetric GARCH model.

The EGARCH is designed to capture both (1) the asymmetric effects between positive and negative shocks on the returns and (2) the effects of big shocks.

The EGARCH[8] model is given by:

$$\ln \sigma_t^2 = \omega + \alpha Z_{t-1} + \gamma(|Z_{t-1}| - \mathbb{E}[|Z_{t-1}|]) + \beta \ln \sigma_{t-1}^2,$$

where $Z_{t-1} = \frac{\varepsilon_{t-1}}{\sigma_{t-1}}$.

[8] Nelson, D. B. (1991), Conditional heteroskedasticity in asset returns: A new approach, Econometrica, 59, 347-370.

GARCH Extensions: EGARCH (continued)

The EGARCH model does not require any restriction on the parameters because, since the equation is on log variance instead of variance itself, the positivity of the variance is automatically satisfied $\forall(\omega, \alpha, \gamma, \beta) \in \mathbb{R}^4$:

$$\sigma_t^2 = \exp(\omega + \alpha Z_{t-1} + \gamma(|Z_{t-1}| - \mathbb{E}[|Z_{t-1}|]) + \beta \ln \sigma_{t-1}^2) > 0$$

The EGARCH model captures the asymmetric effects between positive and negative shocks on the returns, since:

$$\frac{\partial \ln \sigma_t^2}{\partial |Z_{t-1}|} = \begin{cases} \gamma - \alpha & \text{if } Z_{t-1} < 0 \\ \gamma + \alpha & \text{otherwise} \end{cases}$$

The leverage effect, i.e., the fact that negative shocks at time $t-1$ have a stronger impact on the variance at time t than positive shocks, implies that $\alpha < 0$.

GARCH Extensions: EGARCH (continued)

The term $(|Z_{t-1}| - \mathbb{E}[|Z_{t-1}|])$ measures the magnitude of the (positive or negative) shocks.

If the parameter γ is positive, then the big shocks (compared to their expected value) have a stronger impact on the variance than the small shocks.

The mean $\mathbb{E}[|Z_{t-1}|]$ is a constant that depends on the distribution of Z_t. For the Gaussian distribution, we have:

$$\mathbb{E}[|Z_t|] = \sqrt{\frac{2}{\pi}}$$

Fitting SPX Returns with the GJR-GARCH Model

Parameter	Coefficient	Std Error	t-stat	p-value
Mean Model				
Const	0.0250	0.0106	2.362	0.0182
Volatility Model				
ω	0.0210	0.0084	2.508	0.0122
α_1	0.0000	0.0190	0.000	1.0000
γ_1	0.1647	0.0342	4.811	1.503×10^{-6}
β_1	0.8961	0.0362	24.725	5.722×10^{-135}

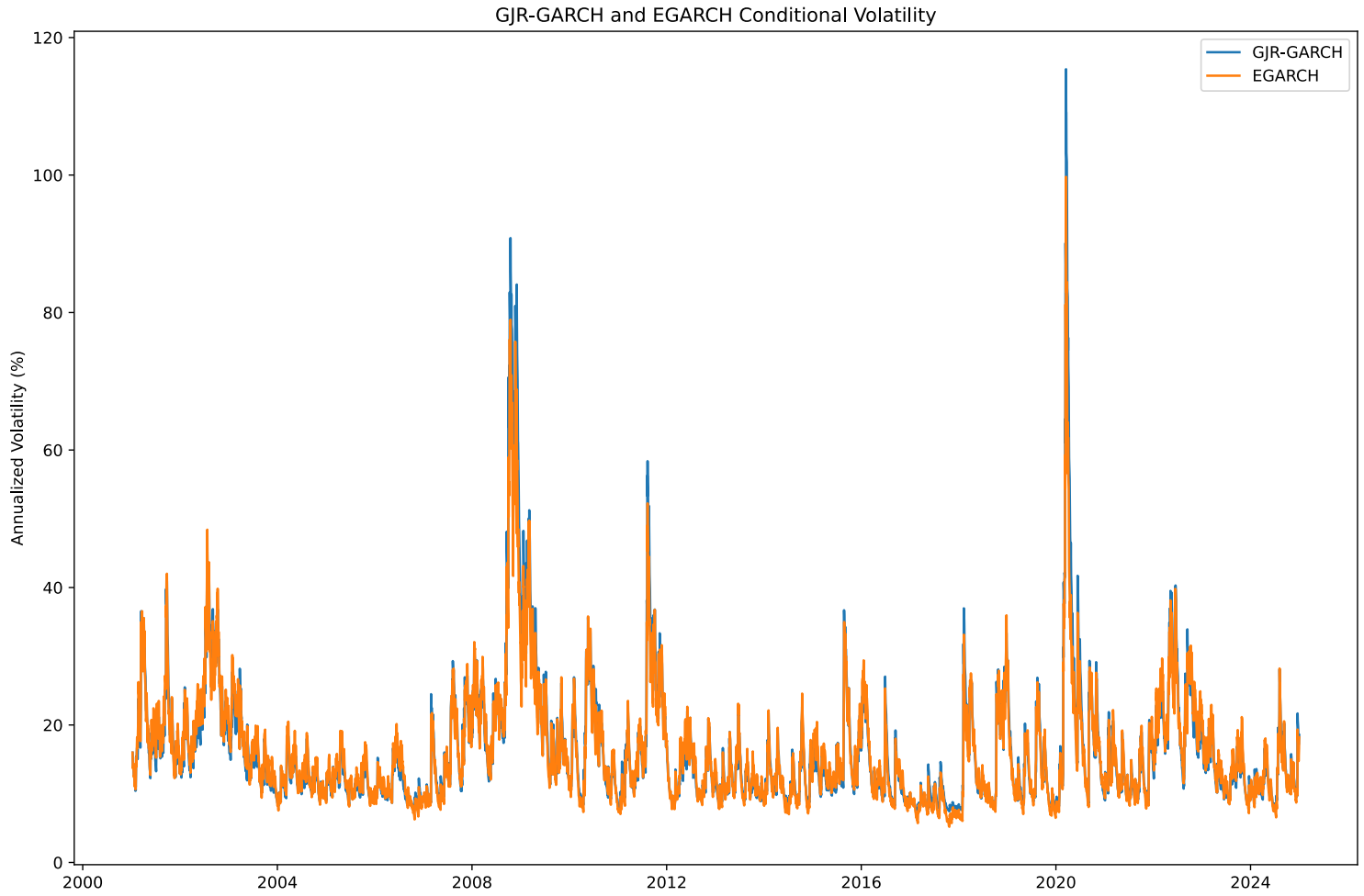
Fitting SPX Returns with the EGARCH Model

Parameter	Coefficient	Std Error	t-stat	p-value
Mean Model				
Const	0.0253	9.972×10^{-3}	2.534	1.127×10^{-2}
Volatility Model				
ω	1.382×10^{-3}	3.120×10^{-3}	0.443	0.658
α_1	0.1555	1.904×10^{-2}	8.170	3.084×10^{-16}
γ_1	-0.1370	1.621×10^{-2}	-8.450	2.917×10^{-17}
β_1	0.9724	4.577×10^{-3}	212.425	0.000

Performance of GARCH Models

Model	AIC	BIC	Log-Likelihood
EGARCH	16187.68	16221.20	−8088.84
GJR-GARCH	16207.67	16241.20	−8098.84
GARCH	16412.35	16439.17	−8202.17

Conditional Volatility of SPX Returns from the GJR-GARCH and EGARCH Models



GARCH Models for Options Pricing

GARCH models have been extensively used to perform option pricing due to its ease of implementation and the flexibility of the model to capture the volatility dynamics of the underlying asset.

Compared to one-period models, GARCH models are able to time-varying dynamics of the volatility and higher moments, which have potential to fit the term structure of implied volatility.

The model construction follows the following steps:

1. Specify the GARCH model under the physical probability measure.
2. Specify the Radon-Nikodym derivative of the risk-neutral probability measure with respect to the physical probability measure.
3. Derive the GARCH model under the risk-neutral probability measure.

Compared to continuous time models, the fitting results are easier to interpret.

The Physical Probability Measure Dynamics

Following the standard GARCH model specified previously, we have:

$$\begin{cases} r_t = \ln(S_t/S_{t-1}) = r + \lambda\sqrt{h_t} - \frac{1}{2}h_t + \sqrt{h_t}Z_t, \\ h_t = \omega + \alpha h_{t-1}Z_{t-1}^2 + \beta h_{t-1}, \end{cases}$$

where $Z_t \sim$ i.i.d.$N(0,1)$ under $\mathbb{P}$.

This fits into the GARCH-M framework, under which we have that

$$r_t = c + \sqrt{h_t}Z_t,$$

where $c = r + \lambda\sqrt{h_t} - \frac{1}{2}h_t$.

The Risk-Neutral Probability Measure Dynamics

There remains the question of how to specify the Radon-Nikodym derivative of the risk-neutral probability measure with respect to the physical probability measure.

We can follow the locally risk-neutral valuation relationship (LRNVR) approach, which is based on the following relationship:

$$\mathbb{E}^{\mathbb{Q}}_{t-1}[e^{r_t}] = e^r, \ \mathrm{Var}^{\mathbb{Q}}_{t-1}(r_t) = \mathrm{Var}^{\mathbb{P}}_{t-1}(r_t) = h_t.$$

Under the LRNVR, the asset return dynamics is given by:

$$\begin{cases} r_t = r - \frac{1}{2}h_t + \sqrt{h_t}Z^*_t, \\ h_t = \omega + \alpha h_{t-1}(Z^*_{t-1} - \lambda)^2 + \beta h_{t-1}, \end{cases}$$

where $Z^*_t = Z_t + \lambda \sim$ i.i.d.$N(0,1)$ under $\mathbb{Q}$.

To Consider Conditional Leverage Effect

We note that the change-of-measure result still holds if we vary the dynamics of h_t.

In general, its dynamics can be written as:

$$h_t = \omega + \alpha h_{t-1} f(Z_{t-1}) + \beta h_{t-1},$$

where $f(Z_{t-1})$ is a function of Z_{t-1}, which may induce asymmetric effect of innovation on the volatility and hence conditional leverage effect.

For example, we can consider the following functions:

- Leverage: $f(Z_{t-1}) = (Z_{t-1} - \theta)^2$.
- News: $f(Z_{t-1}) = (|Z_{t-1} - \theta| - \kappa(Z_{t-1} - \theta))^2$.
- Power: $f(Z_{t-1}) = (Z_{t-1} - \theta)^\gamma$.

To Fit The Parameters

We can fit the parameters of the GARCH model under the physical probability measure using the historical data by maximum likelihood estimation.

The log-likelihood function is given by:

$$\ell(\theta) = -(T-1)\ln(2\pi) - \frac{1}{2}\sum_{t=2}^{T}\ln(h_t) - \frac{1}{2}\sum_{t=1}^{T}\frac{(r_t - r - \lambda\sqrt{h_t} + \frac{1}{2}h_t)^2}{h_t}.$$

The estimated parameters can then be used to price options. In most cases, analytical pricing formula is not availabe, so we can use Monte Carlo simulation to obtain approximate option prices.

In the LRNVR framework, the option price is dependent on the risk-premium λ, which can be estimated from the option data.

The Heston-Nandi Model

The Heston-Nandi model[9] is an affine GARCH model that allows for tractable pricing of options.

$$\begin{cases} r_t = r + \lambda\sqrt{h_t} - \frac{1}{2}h_t + \sqrt{h_t}Z_t, \\ h_t = \omega + \alpha(Z_{t-1} - \theta\sqrt{h_{t-1}})^2 + \beta h_{t-1}, \end{cases}$$

where $Z_t \sim$ i.i.d.$N(0,1)$ under $\mathbb{P}$.

Under this model and the LRNVR, the log price admits closed-form characteristic function.

[9] Heston, S., Nandi, S., 2000. A closed-form GARCH option valuation model. Review of Financial Studies 13, 585-625.